Communicative Efficiency in Adjective Ordering: Between Hierarchical Composition and Incremental Pragmatics

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Abstract

Adjective ordering preferences, such as placing size before colour, have been explained by two accounts of communicative efficiency. Hierarchical composition favours placing less subjective adjectives closer to the noun, where they restrict the interpretation of more subjective ones in the complete utterance; incremental pragmatics favours placing more discriminating adjectives earlier, where each adjective helps the listener narrow down the referent sooner. However, the two accounts can favour opposing orders when the scene makes the less subjective adjective the more discriminating one. Do speakers resolve this conflict by evaluating the complete utterance, or each adjective as they produce it? We develop Rational Speech Act production models that formalise both accounts and span this range, from a global to an incremental speaker. Between these endpoints, a plan-guided speaker chooses each adjective for its immediate usefulness while looking ahead to the complete utterance, so that both pressures can influence adjective order. In German production data, speakers adapted which adjectives they mentioned to scene variation, while their order largely followed a stable convention. We found that the plan-guided speaker best predicted these data, whereas modelling hierarchical composition added no predictive benefit. Simulations nevertheless showed that composition can favour size-first order, but only under restricted conditions. Together, these findings motivate an account of boundedly rational communication in which partial planning supports good-enough production.

Keywords: adjective ordering, communicative efficiency, reference production, Rational Speech Act, Bayesian cognitive modelling

1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with what is informative in a particular scene. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Sproat and Shih, 1991; Trotzke and Wittenberg, 2019; Scontras, 2023). Communicative efficiency has been proposed as an explanation for both stable adjective-order preferences and their sensitivity to referential context (Scontras, Degen, and Goodman, 2017; Franke, Scontras, and Simonic, 2019; Futrell, Dyer, and Scontras, 2020).

But what counts as efficient—and therefore early—for the listener in a complex referential setting? Two accounts give different answers, one based on hierarchical composition and one on incremental pragmatics. Information-theoretic accounts additionally relate ordering to incremental processing under memory limits (Hahn et al., 2018; Futrell, Dyer, and Scontras, 2020; Futrell, 2023). We focus on composition and incremental pragmatics because they make opposite predictions in referential scenes and can be formalised within the same speaker model. The first account, hierarchical composition, starts from the observation that more subjective adjectives tend to occur farther from the noun, which yields a size-first preference for size and colour (Scontras, Degen, and Goodman, 2017; Trainin and Shetreet, 2021). On this account, the noun-adjacent adjective restricts the context in which an outer adjective is interpreted (Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Goodman, 2019; Scontras, Bar-Sever, et al., 2020). In *the big blue sticker*, *blue* combines with *sticker* before *big*, potentially narrowing the comparison class against which size is evaluated. Because colour is less subjective than size, evaluating *big* among the blue objects can reduce uncertainty about its interpretation.

The second account, incremental pragmatics, concerns the order in which the listener receives information. Listeners use partial linguistic input and visual context to restrict candidate referents before a noun phrase is complete (Tanenhaus et al., 1995; Sedivy et al., 1999). Placing the more discriminating property first can facilitate this process, favouring colour-first order when colour identifies the target sooner (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The two accounts therefore conflict when the less subjective property is also the more discriminating one. In a scene where colour alone identifies the

target, composition still favours *the big blue sticker*, whereas incremental pragmatics favours *the blue big sticker*. The compositional benefit of size-first order depends on the complete phrase, whereas the incremental benefit of colour-first order depends on each adjective as it is produced.

However, there is another way of looking at this conflict, and the competition between size-first and colour-first order may never arise in the first place. Whether either order is produced depends first on which adjectives are selected for an utterance. In a colour-sufficient scene, *blue* in *the blue sticker* may complete the utterance alone, so that size and colour need not be ordered at all. Speakers may nevertheless add *big*, even though it is redundant. Redundancy also occurs in size-sufficient scenes, where speakers may produce *the big blue sticker* although size alone identifies the target. Redundant colour, however, is far more common than redundant size, an asymmetry that has been attributed to the higher salience and lower context-dependence of colour (Tarenskeen, Broersma, and Geurts, 2015; Rubio-Fernández, 2016; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). Utterances with both adjectives may therefore arise mostly in size-sufficient scenes, and less often in the colour-sufficient scenes where the two accounts conflict. Adding colour can help the listener, especially when size differences are hard to perceive (Rubio-Fernández, 2016; Degen et al., 2020). This observation motivates the manipulation of size discriminability in our experiment, in which the target stands out more or less clearly as big relative to its comparison class (Section 2).

When speakers do produce both adjectives, previous studies show that the conventional size-first preference is so robust that, even when size is the least discriminating property, it is still preferred first, and reversed orders such as *the blue big sticker* are rare (Danks and Schwenk, 1972; Pechmann, 1989; Schlotterbeck and Wang, 2023). Speakers thus keep a stable order while adapting the content of an utterance to the scene. How the two are combined may depend on how far ahead an utterance is evaluated during production. This scope of evaluation also determines which efficiency benefit can shape the utterance, because the compositional benefit depends on the complete phrase and the incremental benefit on each adjective. Again, in a colour-sufficient scene, a speaker who has planned the complete utterance can begin with *big*, even though *big* is not the most salient adjective at that point, whereas a speaker who evaluates one adjective at a time is likely to begin with *blue*. Which order a speaker produces may thus depend on how much of the utterance is planned in advance.

At one end of this range, a global speaker compares complete utterances as wholes. At the other, a fully incremental speaker evaluates each next adjective against the alternatives available after the words already produced (Cohn-Gordon, Goodman, and Potts, 2019; Waldon and Degen, 2021). Research on bounded planning suggests that human speakers lie between these endpoints, beginning to speak before every part of an utterance has been prepared and continuing to plan during articulation (Ferreira and Swets, 2002; Konopka and Meyer, 2014). A speaker between these endpoints may choose each adjective for its immediate usefulness while looking ahead to the complete utterance, so that both pressures can influence adjective order. We call this intermediate hypothesis *plan-guided production*. A free parameter determines the degree of planning between these endpoints. Planning an utterance involves selecting properties for mention as well as arranging them (Fukumura, Van Gompel, and Pickering, 2010; Van Gompel et al., 2019). Selection and ordering together make the full distribution of one-, two-, and three-adjective utterances the relevant production outcome.

The Rational Speech Act (RSA) framework can predict this distribution by relating speaker choice to an utterance's informativeness and its production cost (Frank and Goodman, 2012; Goodman and Frank, 2016; Degen, 2023). Because both efficiency accounts concern how informative an utterance is, they can be formalised within the same RSA model and compared on the same data. The framework also assigns a probability to every possible utterance, so it extends naturally from two fixed orders to the full set of utterances that speakers produce. Schlotterbeck and Wang (2023) asked participants to rate size-first and colour-first utterances on a slider. Such ratings present both orders for comparison, so they cannot show which properties speakers would mention. Size-first preferences were strongest when size alone identified the target and weakest when colour alone did. A direct replication of this slider study showed the same pattern (Appendix A.1). They also proposed an incremental RSA model that combined the two efficiency accounts, but the model compared only two fixed adjective orders. The separate predictive contributions of the two accounts remained unresolved. It also remained open how they apply when speakers produce the utterances themselves.

Our central research question is: *How do speakers reconcile the pressures from hierarchical semantic composition and incremental pragmatics when selecting and ordering adjectives?* We address this question with two sets of matched model comparisons, each isolating one theoretical assumption while holding all other components constant. The first targets the scope of evaluation that separates the two accounts. It asks which degree of planning best predicts the utterances speakers produce. The second asks whether narrowing the comparison class for size, as proposed by the hierarchical compositional account, improves prediction. We test these models on a German production experiment, in which speakers described a target object in a visual scene. Their utterances show which content, length, and order speakers produce when they formulate an utterance themselves. Methodologically, we extend RSA production models from two fixed orders to the full set of utterances with one, two, or three adjectives

for size, colour, and form. We estimate the models hierarchically across participants and compare them by how well they predict held-out data (Vehtari, Gelman, and Gabry, 2017). A controlled simulation then examines under which conditions hierarchical composition can favour size-first order.

Figure 1 provides an overview of the study, including the approach, the models compared, and the main findings. Section 2 presents the production experiment, followed by the models and their degrees of planning in Section 3. Section 4 presents the predictive model comparisons; Section 5 examines the conditions under which communication favours size-first order. Section 6 discusses the implications for the two efficiency accounts and for bounded planning in production.

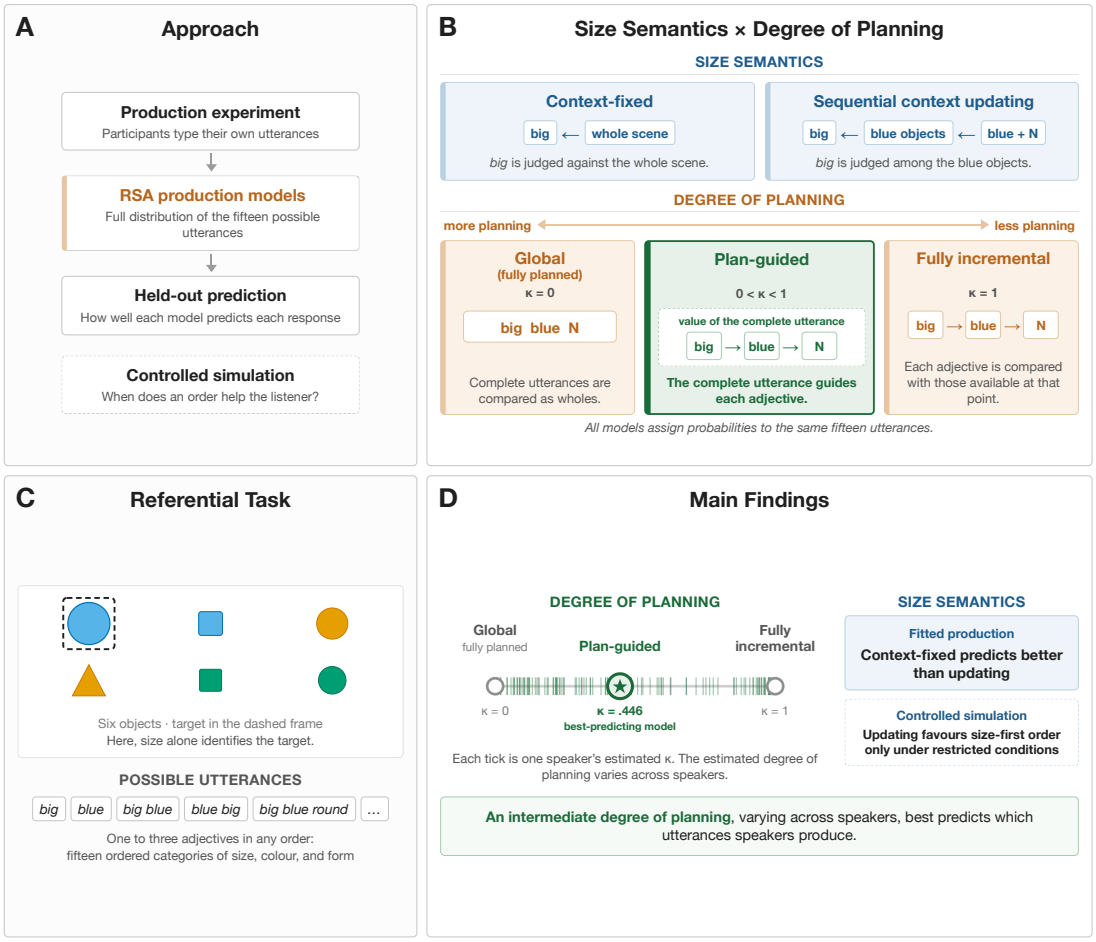


Figure 1: Overview of the approach, models, and findings. (A) RSA production models are fitted to the production experiment and compared by held-out prediction. Controlled simulations examine when an order helps the listener. (B) The production models cross two size semantics with three degrees of planning. The comparison class for *big* is either the whole scene (context-fixed) or the objects matching the noun-adjacent adjective (updating, as proposed by the compositional account). Production ranges from fully planned (global, $\kappa = 0$) through plan-guided ($0 < \kappa < 1$) to fully incremental ($\kappa = 1$). (C) An example display in which size alone identifies the target, with examples of the fifteen possible utterances. (D) Plan-guided production predicts the production data best (estimated $\kappa = .446$); ticks show each participant's estimated κ. Context-fixed semantics predicts better than updating. In the simulations, the listener's size-first preference depends on updating only when two complete utterances are compared.

2 Production Experiment

Whether speakers resolve the conflict between the two accounts by evaluating complete utterances or single adjectives becomes visible only when they select the adjectives themselves. The experiment builds on the slider-rating study of Schlotterbeck and Wang (2023), in which participants compared two phrases that differed only in the order of the size and colour adjectives, such as *den großen blauen Aufkleber* and *den blauen großen Aufkleber*. A

direct replication of the slider study showed the same relation between referential context and adjective-order preference: size-first ratings were strongest when size alone identified the target and weakest when colour alone did, and a small size-first preference remained in colour-sufficient displays (Appendix A.1). The interaction between size discriminability and referential context observed in the original data was not clearly replicated. The present experiment used the same displays and manipulations, but replaced the comparison of the two phrases with direct production. Direct production allows adjective selection and utterance length to vary along with order. As in the slider study, the experiment manipulated referential context within subjects and size discriminability between subjects. Referential context varies whether size, colour, or both are needed to identify the target, manipulating the relative usefulness of the two properties (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Size discriminability separately varies how clearly size differences can be perceived. Crossing these factors allows us to ask whether the influence of size discriminability depends on which properties support reference, and how these pressures affect adjective selection, length, and order.

2.1 Method

2.1.1 Participants

The experiment was conducted online via Prolific, with compensation set slightly above the platform's median level. Participants reported German as their native language. The retained sample comprised 113 participants, 60 in the low- and 53 in the high-discriminability group. Participants from the slider studies were ineligible. Exclusion criteria and the trials retained for each analysis appear in Table 1.

2.1.2 Materials

The task was presented as a reference game about collectible stickers: a partner saw the same stickers in a different order and lacked the target frame, making spatial descriptions uninformative. On each critical trial, participants saw six objects with one framed target (Figure 2, Panel A). The noun was always *Aufkleber* ('sticker'), and the materials varied the objects' size, colour, and form across counterbalanced lists. The target was always the largest object of its colour and form, so it could always be described as *big*. Stimulus construction and trial counts appear in Appendix A.

2.1.3 Design

The experiment used a 3 (referential context: first property sufficient, second property sufficient, both necessary; within-subjects) $\times$ 2 (size discriminability: high, low; between-subjects) mixed design. Each context was realised with three adjective pairs (size–colour, size–form, colour–form), which determined the two properties that could distinguish the target. Referential context changed which properties were needed to identify the target; in size–colour trials, for example, size alone (size sufficient), colour alone (colour sufficient), or both properties (both necessary) identified the target. Target–competitor size differences were more pronounced in the high-discriminability group than in the low-discriminability group (Figure 2). The behavioural analyses focus on size–colour trials, the classic contrast in the ordering literature and our motivating example. The production models are fitted to all three adjective pairs, with colour–form trials providing a baseline without a subjective adjective. The two experimental hypotheses concern the interaction between the two factors. Lower size discriminability should increase redundant adjective use more strongly when size alone is sufficient, and higher discriminability should increase size-initial responses more strongly when both properties are necessary.

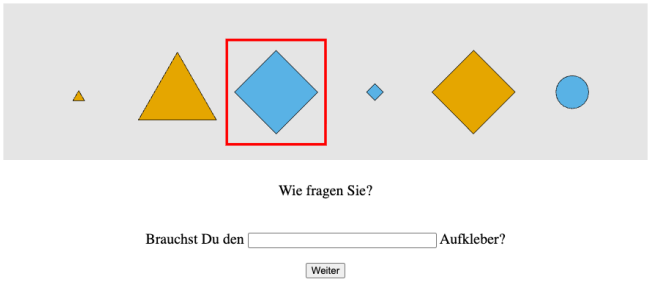
2.1.4 Procedure

Participants saw the framed target together with the fixed sentence frame *den ___ Aufkleber* and typed the adjective material before the noun. This response format allowed us to observe which properties they mentioned, how many adjectives they used, and how they ordered them. Practice trials introduced the relevant colour and form vocabulary, and main-trial order was randomised within blocks.

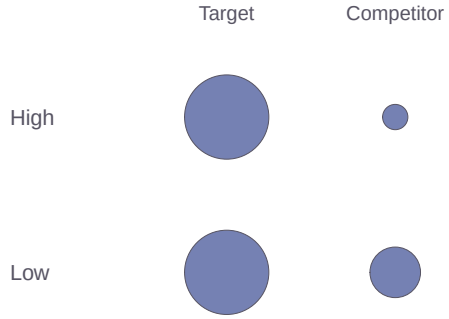
2.1.5 Response Coding and Analysis

Responses were annotated following guidelines that preserve adjective content and order. A production response could contain a single adjective, such as *blue*; two adjectives, such as *big blue*; or three, such as *big blue round*. Coding each adjective as size (S), colour (C), or form (F) gives fifteen ordered categories: three single adjectives, six two-adjective sequences, and six three-adjective sequences. All production models assign probabilities to these

A



B



C

Design dimension	Levels
Referential context	First property sufficient; second property sufficient; both necessary (within-subjects)
Size discriminability	High; low (between-subjects)
Adjective pair	Size–colour; size–form; colour–form (stimulus sets, crossed with referential context)

Figure 2: Experimental design and materials. Panel A shows a trial screenshot from the production experiment: the framed object is the target, and participants typed the adjective material into the sentence frame. Panel B illustrates the size-discriminability manipulation schematically using recorded stimulus levels: the target–competitor size gap is larger under high discriminability and smaller under low discriminability. Panel C summarises the 3×2 design and the three adjective pairs.

fifteen categories. The complete annotation scheme, including the treatment of responses outside these categories, appears in Appendix A.

The annotated responses were further classified for the two experimental hypotheses. *Redundant adjective use* indicates a response containing more properties than needed for unique reference: multiple adjectives in a one-property-sufficient context, or three adjectives when both properties are needed. *Size-initiality* indicates that a response begins with size, including a response consisting of size alone. It reflects adjective selection as well as order. Order conditional on producing both size and colour is described separately. The two outcomes were analysed with Bayesian hierarchical models containing referential context, size discriminability, and their interaction; the full specifications and planned contrasts appear in Appendix A.

2.2 Results

Adjective content and length. The observed response distribution indicated that referential context shaped adjective content and utterance length (Figure 3, Panel A). Both-necessary scenes generally elicited longer utterances, whereas one-property-sufficient scenes more often elicited a single adjective. Colour-sufficient scenes were commonly resolved with colour alone, so no size–colour ordering decision was observed on those trials. Production also revealed substantial use of three-adjective utterances.

Redundant adjective use. When size alone identified the target, most responses still contained more than one adjective (Figure 3, Panel B). Such redundant utterances were more frequent when size differences were hard to see, by 23.4 percentage points under low discriminability (95% CrI $[15.1, 31.8]$; posterior probability $> .999$). In both-necessary and colour-sufficient displays, discriminability made a smaller difference, and the credible intervals included zero. The Bayesian hierarchical analysis confirmed that the effect of discriminability was larger when size alone was sufficient than on average in the other two contexts (difference $= 15.3$ percentage points, 95% CrI $[7.2, 23.6]$; posterior probability $> .999$ in the hypothesised direction).

Initial adjective and conditional order. Speakers began with size most often when size alone identified the target, less often when both properties were needed, and least often when colour alone did (Figure 3, Panel C). Both differences were credible (19.3 percentage points, 95% CrI $[13.9, 24.8]$; 53.1 percentage points, 95% CrI $[45.9, 60.1]$). Size discriminability mattered mainly when both properties were needed (difference from the one-property-sufficient contexts $= 16.9$ percentage points, 95% CrI $[7.7, 25.6]$; posterior probability $> .999$ in the

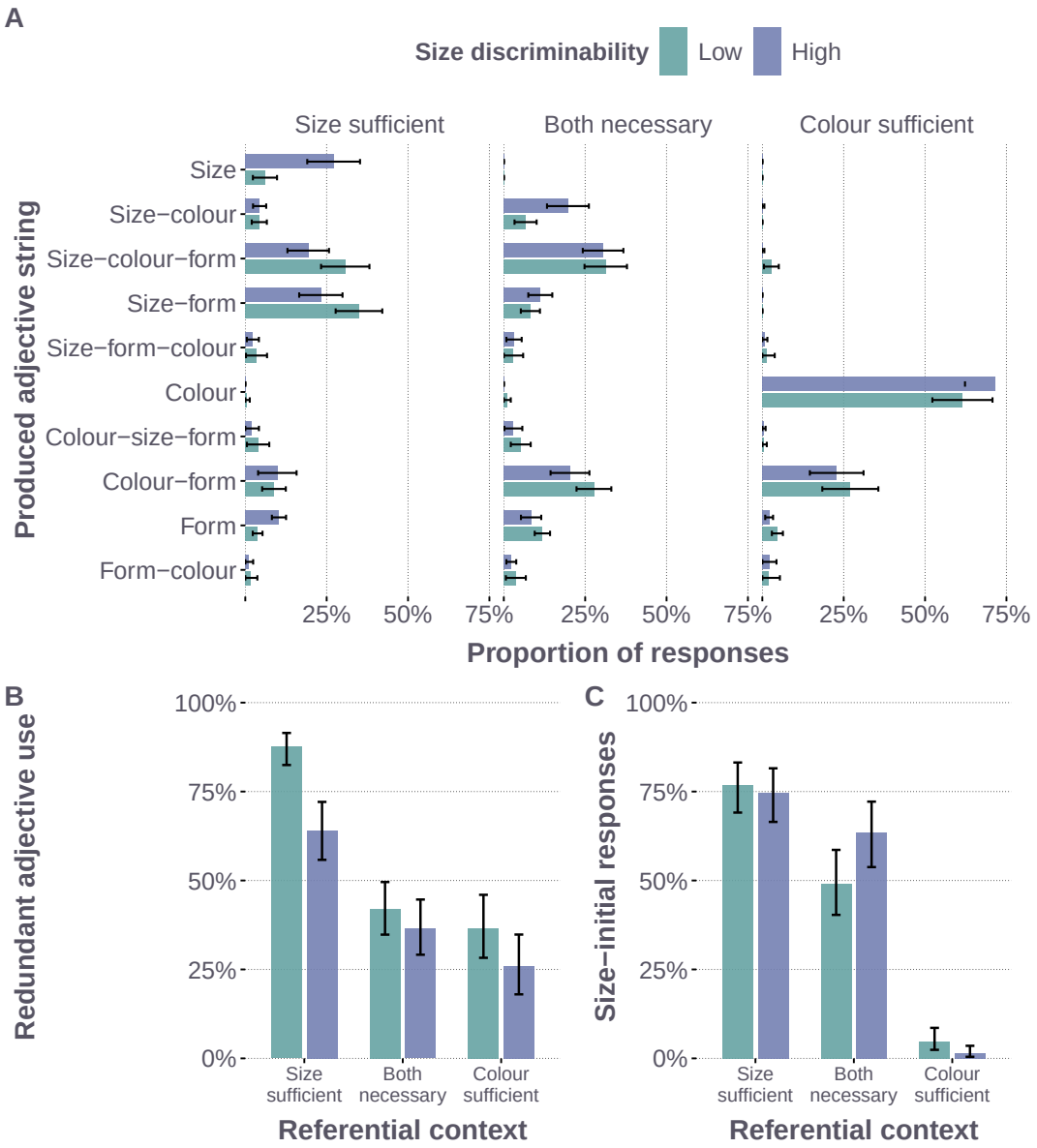


Figure 3: Size–colour production responses by referential context and size discriminability. Panel A shows the ordered response distribution, with bars for mean participant-level proportions and 95% confidence intervals. Adjective labels preserve surface order, and the panel shows responses reaching at least 2% in one condition. Panel B shows redundant adjective use, defined as a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel C shows the probability that a response begins with size, which combines adjective selection and order. In Panels B and C, bars show Bayesian posterior medians and error bars show 95% credible intervals.

hypothesised direction). In these displays, speakers began with size more often when size differences were easy to see (14.1 percentage points, 95% CrI [3.6, 24.2]; posterior probability .997). In the two one-property-sufficient contexts, discriminability had at most a small effect on size-initial responses. Conditional on producing both size and colour, size-first utterances remained more frequent, and colour-first utterances were comparatively rare.

2.3 Discussion

The results supported the hypothesis that reduced size discriminability particularly increases redundant adjective use when size alone can identify the target. They also supported the hypothesis that higher size discriminability particularly favours size-initial responses when both properties are needed to identify the target. Speakers were more likely to begin with size when it was more useful for reference, including when they used size as the only adjective. The conditional size-first preference describes responses containing both properties, while size-initiality also includes the decision to use size alone. Referential context also affected how often speakers produced the two adjectives whose order the two efficiency accounts compare.

Three-adjective responses extend beyond the material required for unique reference and involve the selection and ordering of several adjectives. Modelling these data requires assigning probability to utterances of different lengths and contents, including redundant utterances. The following framework represents these alternatives within a common utterance space, allowing us to compare how hierarchical composition and the degree of planning contribute to their use.

3 A Plan-Guided Production Model

In the colour-sufficient scene of the introduction, a fully planned speaker compares complete utterances, such as *blue*, *big blue*, and *blue big*, whereas a fully incremental speaker evaluates the first adjective on its own. Both speakers pursue the same goal of identifying the target; they differ only in how far ahead they look. Which of them do human speakers resemble, or do they lie in between?

The Rational Speech Act (RSA) framework models a speaker who selects utterances according to their informativeness about the intended referent and their production cost (Frank and Goodman, 2012; Goodman and Frank, 2016). Standard RSA speakers evaluate complete utterances, whereas incremental variants evaluate one word at a time (Cohn-Gordon, Goodman, and Potts, 2019). Our model keeps this measure of informativeness but treats the choice between these two forms of evaluation as a matter of degree, estimated from the data. Because planning cannot be observed directly in typed utterances, the model infers it from the distribution of utterances that speakers produce: which properties they mention, how many adjectives they use, and how they order them. The models also differ in whether the comparison class for size is updated as adjective meanings compose from the noun outwards. Panel B of Figure 1 shows these two dimensions.

3.1 Literal Interpretation and Meaning Function

All models use the observed six-object display. Size has a graded meaning: an object counts as *big* to the degree that it is larger than the other objects in its comparison class (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019). Colour meanings are graded as well: objects of the named colour receive most, but not all, of the weight. Form is treated as uninformative in the literal semantics; speakers' use of form is captured by salience and by the length and selection terms introduced below. These semantic settings are shared across the production architectures.

For every complete or partial adjective string, the literal interpretation L composes the adjectives hierarchically, applying modification from the noun outwards. L corresponds to the literal listener of RSA models and serves here as a semantic layer rather than a separate agent. For example, in *big blue sticker*, *blue* modifies *sticker* before *big* is interpreted. Let $c_1, \ldots, c_T$ denote the modifiers of a string in this order, so that c_1 is the adjective next to the noun. Starting from a uniform prior over the objects, $C_0 = P_0$, composition updates a distribution over objects at each step:

$$C_j(o) \propto C_{j-1}(o)\, [\![c_j]\!]_{\rho_j}(o), \qquad j = 1, \ldots, T, \tag{1}$$

where $[\![c_j]\!]_{\rho_j}$ is the modifier's meaning under the selected semantic regime. The final state gives the literal interpretation of the string as a distribution over objects.

Hierarchical composition affects order only if an inner adjective can change how an outer adjective is interpreted. The two semantic regimes differ in exactly this respect. Under *context-fixed* semantics, the comparison class for size remains the whole scene. Under *sequential context updating*, the comparison class is recomputed

after each modifier, so that in *big blue*, *big* is judged among objects weighted by how well they match *blue*. Because colour meanings are graded, this restriction is partial. When it makes the target more distinctive in size, updating can make size-first order more informative. The threshold construction and semantic parameters appear in Appendices B and F.

Interpretation proceeds from the noun outwards, but production proceeds from left to right. The speaker evaluates the information available after each adjective in the surface sequence. Let $u_{\leq t} = w_1 \dots w_t$ be the prefix comprising the first t adjectives, and $P_0(o \mid X) = 1/6$ the object prior in display X. Each prefix is interpreted from this prior, applying each of its modifiers once in noun-outward order. For *big blue*, the first prefix is *big* alone, and the second is the composed meaning of *big blue*. Under context-fixed semantics, the interpretation of a prefix is

$$L(o \mid u_{\leq t}, X) \propto P_0(o \mid X) \prod_{j=1}^{t} [\![w_j]\!]_{\text{fix}}(o). \tag{2}$$

Under updating, the comparison class for *big* changes as the prefix grows from *big* to *big blue*. The probability that this interpretation assigns to the target is the communicative value of the prefix.

3.2 Production Architectures

The three production architectures share a *prefix utility*, which accumulates how well the adjectives in an utterance identify the target at each adjective position. This accumulation means that an utterance gains value when it identifies the target early, not only when it is complete. The production architecture determines how this accumulated utility is compared with alternatives: across complete utterances, or among the adjectives available at each point. At position t, the utility of candidate adjective a for target o^* is

$$U_t(a \mid u_{<t}, o^*, X) = \alpha \log L(o^* \mid u_{<t} + a, X) + \lambda_{\text{salience}} z_a, \tag{3}$$

where the pragmatic-optimality parameter α controls sensitivity to informativeness and z_a represents the visual salience of the candidate property. The two terms capture what can make an adjective attractive as soon as it is produced: it discriminates the target, or it is visually salient. For a completed utterance u, the model combines the accumulated utility of its selected adjectives with a term for competition among alternatives at each adjective position:

$$U^{\text{complete}}(u, o^*, X) = \sum_t U_t(w_t \mid u_{<t}, o^*, X), \tag{4}$$

$$\mathcal{N}^{\text{local}}(u, o^*, X) = \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}, o^*, X), \tag{5}$$

where $\mathcal{A}(u_{<t})$ contains the adjective types still available after the current prefix. The complete-utterance utility U^{complete} is the sum of the utilities at each adjective position and is shared by all three production architectures. The local-normalisation term $\mathcal{N}^{\text{local}}$ records the utility of the competing adjectives at each position. When adjectives compete at a given point, a selected adjective receives lower probability if other available adjectives are also useful there. For example, *big blue* and *blue big* contain the same properties but face different competitors at the first position. When colour alone identifies the target, *blue* is a strong competitor, so *big* is less likely to come first. Two completed utterances with similar U^{complete} can receive different probabilities because they pass through prefixes with different degrees of competition. These two terms correspond to the two influences in plan-guided production: the value of the complete utterance and the competition among adjectives at each point.

The probability of a completed utterance is

$$S_\kappa(u \mid o^*, X) \propto \exp\left\{U^{\text{complete}}(u, o^*, X) - \kappa \mathcal{N}^{\text{local}}(u, o^*, X) + \beta \Omega(u) + T(u, X)\right\}. \tag{6}$$

The parameter κ determines how strongly competition among the adjectives available at each point contributes to the probability of a completed utterance. With $\kappa = 0$, this competition drops out, and the *global* speaker compares only the fifteen complete utterances. With $\kappa = 1$, the *fully incremental* speaker produces each adjective by comparing it with the adjectives available at that point, without looking ahead; the probability of an utterance then combines these local choices with the shared terms below.[1] Intermediate values define the *plan-guided* speaker, in which both comparisons contribute. Lower values of κ therefore correspond to a greater degree of planning.

[1] This contrast corresponds to that between locally and globally normalised sequence models in natural language processing (Lafferty, McCallum, and Pereira, 2001; Andor et al., 2016).

In the colour-sufficient scene, κ thus governs whether a speaker is drawn to *blue*, the most useful adjective at that point, or can still begin with *big* as part of a planned utterance. Even the global speaker still evaluates each prefix; it differs only in comparing complete utterances.

Besides informativeness and salience, the utterances are shaped by tendencies that are not specific to planning: speakers follow conventional orders, avoid unnecessarily long utterances, prefer a single adjective when it suffices, and add properties when size is hard to see. The slider replication, for example, showed a small size-first preference even when colour alone identified the target (Appendix A.1). Our account does not aim to derive these tendencies, but a model that ignored them would misattribute their effects to the degree of planning. Following common practice in Bayesian cognitive modelling, we include them as auxiliary components that link the model to the task (Lee and Wagenmakers, 2013; Franke, 2016): a fixed order score $\Omega(u)$ from a German language model, weighted by β, and length and selection terms $T(u, X)$. The order score reflects conventions learned from language use, which may themselves have been shaped by efficiency pressures over time (Franke, Scontras, and Simonic, 2019). Their parameters are estimated jointly with κ, so uncertainty about them is carried into the inference about planning. Because these components are identical across architectures, the architectures differ only in how utterances are evaluated. Their specification appears in Appendix B.

4 Model Comparison and Predictive Evidence

The model comparison tests how hierarchical semantic composition and the degree of planning contribute to the observed distribution of utterances. Crossing the three production architectures with the two semantic regimes gives six matched models (Figure 1, Panel B), which share all other components. Two further models refine the plan-guided architecture with participant-specific κ_i under each semantic regime. All also allow participant variation in pragmatic optimality α_i and stable-order weight β_i. The comparison asks whether an intermediate degree of planning improves prediction over both endpoints, whether updating the size comparison class adds further predictive value, and whether allowing the degree of planning to vary across speakers improves prediction. We then examine which production patterns the best-predicting model captures.

4.1 Inference and Predictive Comparison

We fitted the production models to the 9,100 annotated responses from the 113 participants in the production experiment, including size–colour, size–form, and colour–form trials. Each response was represented as one of the fifteen ordered adjective categories used by every model. Bayesian inference was performed in NumPyro (Phan, Pradhan, and Jankowiak, 2019).

Approximate leave-one-out expected log predictive density (ELPD) measures how much probability each model assigns to held-out responses (Vehtari, Gelman, and Gabry, 2017). Higher ELPD indicates better prediction of the complete utterance distribution. Uncertainty in paired model comparisons is estimated with a participant-level Bayesian bootstrap and reported as 95% credible intervals. As a robustness check, a ten-fold cross-validation over participants predicted each held-out participant as a new speaker and led to the same conclusions for the architecture, semantic, and participant-variation comparisons reported below (Appendix C.5). In total, the comparisons rest on 66 fitted production models: the eight models above, 50 cross-validation fits, and eight refits for the robustness checks in Appendix C.5. The inference procedure and numerical diagnostics appear in Appendix C.

4.2 Production Architecture

For the question of which degree of planning best predicts production, the Bayesian model comparison favoured plan-guided production over both global and fully incremental production (Figure 4, Panel A). Averaged across semantic regimes, the plan-guided model had higher predictive accuracy than the global model ($\Delta\text{ELPD} = 256.9$, 95% CrI $[71.8, 404.7]$) and the fully incremental model ($\Delta\text{ELPD} = 548.5$, 95% CrI $[349.8, 795.1]$). These contrasts supported the plan-guided production hypothesis that the value of the complete utterance and local adjective competition jointly shape the response distribution. Both contrasts held when the two condition-specific length and selection terms were removed from all models (Appendix C.5). In the context-fixed model, the posterior mean of the shared κ was .446, with a 95% CrI of $[.407, .486]$ (Panel B). The estimate lay between the global and fully incremental endpoints.

Under context-fixed semantics, the plan-guided advantage was credible in all three adjective families and largest in size–colour trials ($\Delta\text{ELPD} = 202.4$, 95% CrI $[169.3, 237.2]$; Appendix C.5). Colour-initial responses accounted for 81.7% of the plan-guided model's ELPD advantage over the average of the global and fully incremental models. In the model, for example, once *blue* has been produced, size and form compete for the next

position, and this competition differs from that faced by an utterance beginning with *round*. Different opening adjectives leave different continuations available, allowing competition at later positions to affect the probabilities of the completed utterances.

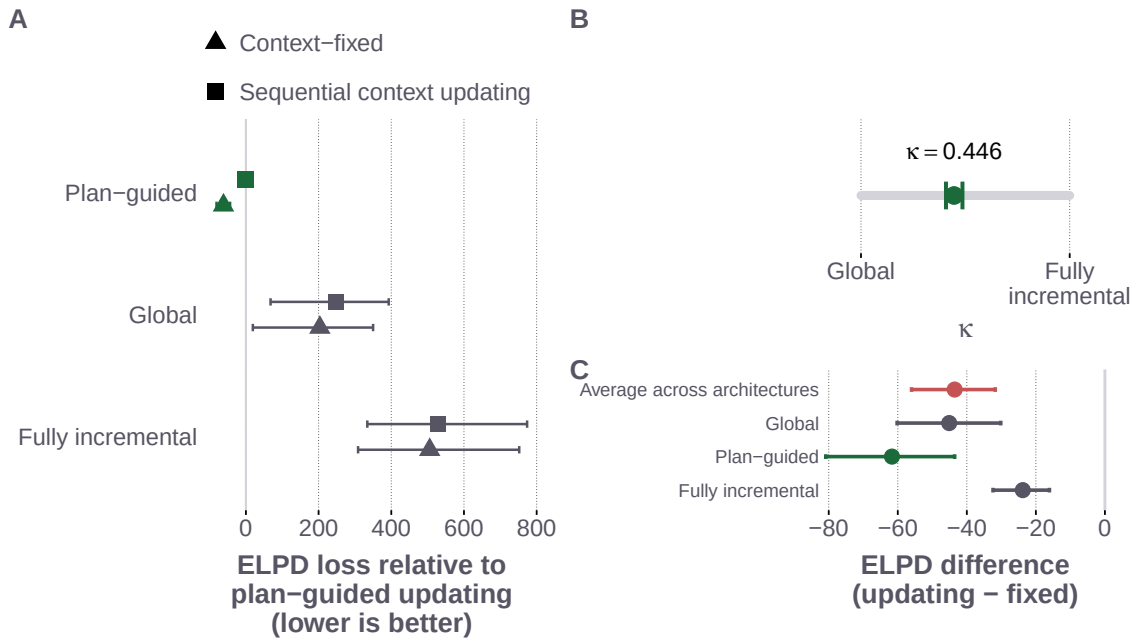


Figure 4: Matched comparisons of production architecture and sequential context updating. Panel A reports predictive loss for all six matched models relative to the plan-guided model with sequential context updating; lower values indicate better leave-one-out prediction. Panel B places the shared κ between the global and fully incremental endpoints. Panel C shows updating-minus-fixed ELPD for each production architecture and their average. Intervals in Panels A and C are 95% credible intervals from the participant-level Bayesian bootstrap; Panel B shows the 95% posterior credible interval for κ. The vertical line in Panel C marks zero.

4.3 Plan-Guided Production and Participant Variation

Allowing κ to vary across participants yielded a credible improvement over the otherwise matched model with a shared κ. Under context-fixed semantics, the participant-specific model gained 364.7 ELPD (95% CrI $[248.2, 521.0]$); the gain under sequential context updating was similar. The posterior mean population location remained intermediate between the global and fully incremental endpoints ($\mu_\kappa = .472$, 95% CrI $[.337, .618]$), alongside substantial variation in participant-specific κ_i. The participant-specific model also predicted better than a benchmark that assigns each participant the better-predicting of the two endpoint models ($\Delta\text{ELPD} = 249.7$, 95% CrI $[103.8, 416.1]$), so its gain does not reduce to a mixture of global-like and fully incremental participants (Appendix C.5).

The predictive benefit was concentrated in two- and three-adjective utterances, whereas predictions of single-adjective responses became less accurate. The gains for longer utterances outweighed this loss and extended to three-adjective utterances, which contain redundant information in the present task. Appendix D reports the response-length contrasts and participant-level diagnostics.

4.4 Hierarchical Semantic Composition

We next asked whether narrowing the comparison class for size, as proposed by the hierarchical compositional account, improves prediction. The comparison favoured context-fixed semantics over sequential context updating under all three production architectures (Figure 4, Panel C). The average updating-minus-fixed contrast was negative ($\Delta\text{ELPD} = -43.5$, 95% CrI $[-56.0, -31.8]$), indicating a credible predictive advantage for the context-fixed models. This advantage remained in the participant-specific plan-guided comparison. Context-fixed semantics also predicted better when colour meanings were made more reliable (updating minus fixed $\Delta\text{ELPD} = -45.9$, 95% CrI $[-59.9, -32.0]$), although both semantic regimes then predicted worse overall (Appendix C.5).

Decomposing the participant-specific model comparison showed that the semantic difference chiefly concerned how much speakers said and which properties they mentioned. Context-fixed semantics improved predictions of utterance length ($\Delta\text{ELPD} = 21.5$, 95% CrI $[12.4, 30.9]$) and adjective content conditional on length ($\Delta\text{ELPD} = 31.1$, 95% CrI $[23.5, 39.2]$). For order conditional on adjective content, the estimated advantage was small and its interval included zero ($\Delta\text{ELPD} = 1.5$, 95% CrI $[-1.8, 4.9]$).

Full contrasts and the score decomposition appear in Appendix C.3.

4.5 Predictive Adequacy of the Best-Predicting Model

Posterior predictive checks assessed how closely the model reproduced the observed context sensitivity of adjective selection, utterance length, and initial position. The plan-guided model with context-fixed semantics and participant-specific κ captured the broad context-dependent shifts in size-initial responses and redundant adjective use (Figure 5, Panel A). The comparison across all fifteen utterances and eighteen experimental conditions showed correspondence between observed and predicted proportions across response cells (Panel B).

The supplementary condition-wise checks made these patterns more explicit (Figure 12). In size–colour trials, the model reproduced frequent colour-only responses when colour was sufficient and the greater use of size alone when size was sufficient and highly discriminable. The main remaining discrepancy concerned the balance between some two- and three-adjective utterances. Descriptive measures of fit appear in Appendix C.

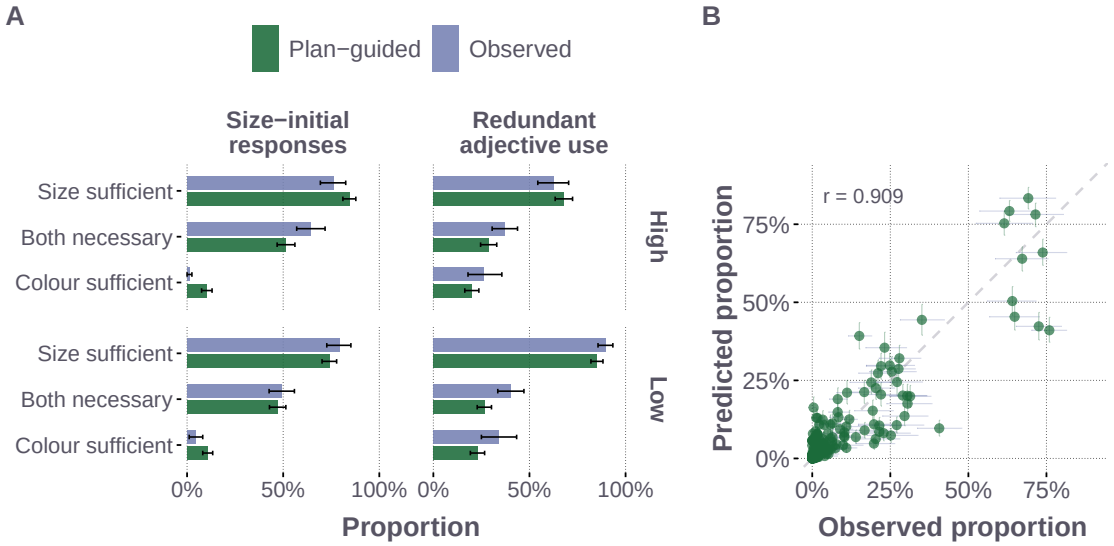


Figure 5: Posterior-predictive adequacy of the best-predicting plan-guided model. Panel A compares observed and posterior-predicted size-initial responses and redundant adjective use in size–colour trials, with referential context on the vertical axis and size discriminability in facet rows. Intervals show participant-bootstrap 95% intervals for the observed proportions and 95% posterior-predictive intervals for the model predictions. Panel B compares all 270 response cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality.

The predictive evidence thus locates the resolution of the conflict in how far ahead utterances are evaluated. Plan-guided production predicted best, whereas narrowing the comparison class for size did not improve prediction. The predictive comparisons do not show whether either semantic regime favours size-first order communicatively. The following simulation examines when such a preference arises and how it changes when adjective selection and utterance length enter the speaker's alternatives.

5 Simulation: Conditions for a Size-First Preference

A communicative account of adjective ordering seeks to explain why one order can be more useful for identifying a referent than another. Earlier simulations found that, on average, reference favours orders that place more subjective adjectives farther from the noun, motivating a pressure that could gradually stabilise ordering preferences across repeated language use (Franke, Scontras, and Simonic, 2019). Sequential context updating provides a mechanism for this preference: the noun-adjacent adjective restricts the context in which an outer, more subjective

adjective is interpreted. The production findings place this proposal within a broader set of choices, because speakers also decide which adjectives to include and how much to say. We therefore ask how the ordering preference changes when the speaker's utility includes information at each adjective position and the speaker can produce utterances of different lengths.

5.1 Controlled Changes in Utility and Available Utterances

The simulation reuses the same 24,000 randomly generated displays across all comparisons. Displays vary in object count and size distribution; the largest object is the target and is assigned blue. Varying the number of blue distractors changes how strongly colour restricts the comparison class for size. The scene-generation procedure and semantic parameter grid appear in Appendix G.

We introduce the production model's assumptions in controlled steps (Figure 6, Panel A). The first comparison gives a global speaker two alternatives, *big blue* and *blue big*, scored by the literal target probability after the completed utterance. The second retains global production ($\kappa = 0$) but introduces the prefix utility in Section 3.2, summing literal log probabilities of the target at each adjective position. For *big blue*, the utility then includes the information supplied by *big* as well as by the completed utterance. Holding global production fixed isolates the contribution of this prefix information. We next add the single-adjective alternatives *big* and *blue*, and finally expand to the fifteen utterances used in production. These four comparisons average over the same grid of 125 semantic parameter settings under each semantic regime.

A further comparison holds the fifteen utterances and production semantic constants fixed while varying the contribution of local competition: global ($\kappa = 0$), plan-guided ($\kappa = .5$), and fully incremental ($\kappa = 1$) production (Panel B). All three architectures use the prefix utility. Throughout these comparisons, pragmatic optimality is $\alpha = 1$, order, salience, length, and selection terms are zero, and final Laplace smoothing is omitted. Appendix G.1 gives the full specification.

The pragmatic listener L^{prag} applies Bayes' rule to the speaker distribution, using an equal prior over objects. An object receives more probability when a speaker intending that object would be more likely to produce the observed utterance. For each context, the size-first preference compares the listener's target probability after the two orders:

$$\Delta_{\text{SF}} = L^{\text{prag}}(o^* \mid \text{“big blue”}) - L^{\text{prag}}(o^* \mid \text{“blue big”}). \tag{7}$$

Positive values favour size-first order. We call this measure the listener-based size-first preference, to distinguish it from the order preferences that speakers showed in production. Other available utterances can affect this measure because the listener reasons about the speaker's choice among them.

5.2 Conditions under which an Order Asymmetry Emerges

With global production and the terminal utility, sequential context updating generated a mean size-first preference of approximately 1.1 percentage points, while context-fixed semantics yielded no difference between the orders (Figure 6, Panel A). With fixed meanings, the two adjective orders have the same completed interpretation and therefore the same target probability under this utility. Updating breaks this symmetry because colour restricts the comparison class for size. This comparison recovers the proposed communicative contribution of hierarchical composition.

Including information available at each adjective position changed the result. With only the two orders available, the global production model yielded a mean colour-first preference under both semantic regimes: approximately 2.7 percentage points with fixed meanings and 1.8 with updating. The first adjectives can differ in informativeness even when the completed meanings coincide, allowing an ordering asymmetry under fixed semantics. Its direction, however, depends on the alternatives available for reference. Adding the single-adjective responses changed the mean preference to size-first under both regimes, to approximately 2.3 and 2.8 percentage points, respectively. The speaker can now identify a referent with one adjective, changing how probability is distributed over the two-adjective utterances and how the pragmatic listener interprets their use. Adjective selection and utterance length thus affect the communicative value of an order even when the two orders themselves remain unchanged.

Expanding to fifteen utterances retained the mean size-first preference, at approximately 2.6 percentage points with fixed semantics and 3.1 with updating. At the semantic constants used in production, all three production architectures generated a mean size-first preference under both semantic regimes (Figure 6, Panel B). The preference was approximately 1.9 percentage points for global production, 1.5 for plan-guided production, and 1.0

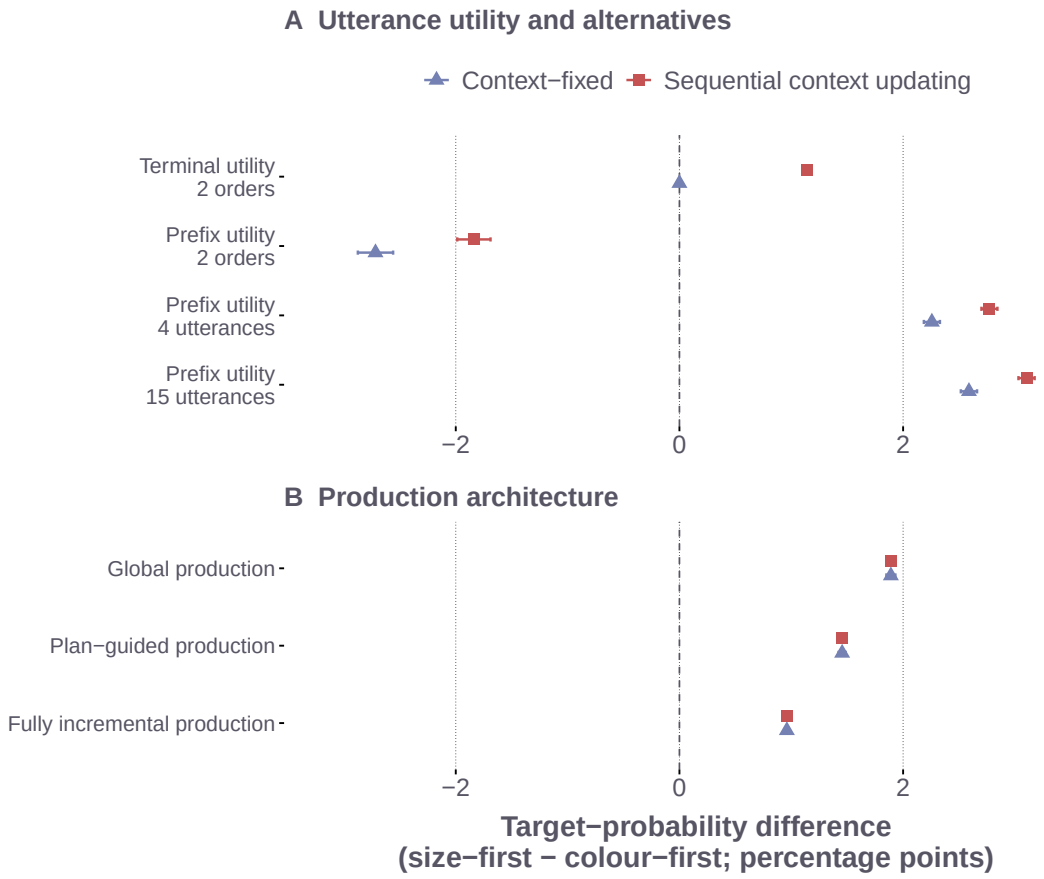


Figure 6: Size-first preference in pragmatic target recovery across utterance utility, alternatives, and production architecture on the same 24,000 random scenes. Panel A holds global production fixed and averages over the semantic parameter grid. The terminal utility uses only the final literal target probability; the prefix utility sums literal log probabilities at each adjective position. The four-utterance space adds single adjectives to the two orders; the fifteen-utterance space contains all production responses. Panel B holds the fifteen utterances and production semantic constants fixed while comparing global ($\kappa = 0$), plan-guided ($\kappa = .5$), and fully incremental ($\kappa = 1$) production. All three use the prefix utility. Points show scene-averaged target-probability differences; positive values favour size-first and negative values favour colour-first. Intervals are ± 1.96 Monte Carlo standard errors, calculated after averaging parameter settings within scenes.

for fully incremental production, with nearly identical values for fixed and updating semantics within each architecture. Increasing the contribution of local adjective competition reduced the magnitude of the preference while preserving its direction.

The simulation shows how the interpretation and selection of adjectives determine which order communication favours. Updating can favour size-first order between two complete utterances, but once single adjectives are available, the option of a shorter utterance changes how the listener interprets the same two-adjective orders.

6 General Discussion

The present findings connect adjective ordering to the cognitive organisation of reference production. Stable size-first preferences coexist with context-sensitive selection of descriptive content and length. Among the matched production models, the best predictions combine evaluation of complete utterances with continued influence from local adjective competition. Participant-specific κ further improves prediction, particularly for utterances containing several adjectives. These findings motivate a plan-guided account of production in which speakers anticipate the communicative value of a complete utterance. For the question of how speakers reconcile hierarchical composition and incremental pragmatics, the results point to the scope of evaluation rather than to the interpretation of size. In the plan-guided model, the two efficiency pressures mainly shaped which adjectives were mentioned

and which adjective came first, whereas the order of size and colour, once both were mentioned, largely followed a stable convention. In the scenes where the two accounts conflict, speakers mostly produced colour alone, so that no ordering of size and colour was observed; when they mentioned both properties, they still placed size first. In the plan-guided model, both patterns are consistent with local competition favouring a colour-only utterance and the stable order term maintaining size-first order in longer utterances.

6.1 Bounded Planning and Redundant Utterances

A useful utterance must be selected from alternatives that differ in content, length, and order. As the number of available properties increases, their combinations and orderings grow combinatorially, making exhaustive comparison increasingly demanding. The cognitive problem is how speakers preserve sensitivity to the communicative consequences of an utterance while limiting the alternatives and continuations they evaluate. Computational tractability constrains the processes and representations that can support this behaviour (Rooij, 2008); approximation alone provides no general guarantee of tractability (Kwisthout, Wareham, and Van Rooij, 2011).

Plan-guided production motivates a hypothesis about how planning could be organised under these constraints. An utterance-level plan could specify properties relevant to identifying the target, constraining the alternatives considered as each adjective is produced. Local choices could remain sensitive to the developing utterance and the properties still available for mention. This organisation accords with evidence that conceptual planning and linguistic formulation can proceed with different scopes (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014). It also connects to good-enough processing, in which representations can remain partial while serving the current task (Ferreira, K. G. D. Bailey, and Ferraro, 2002).

Redundant utterances provide a case in which planning and communicative goals interact. A property may be selected because it becomes accessible early, because it helps the listener identify the target, or because both pressures favour its inclusion (Pechmann, 1989; Engelhardt, K. G. Bailey, and Ferreira, 2006; Rubio-Fernández, 2016; Degen et al., 2020). Readily available information can guide visual search or hedge uncertainty about another property (Rubio-Fernandez, 2019; Tourtouri, Delogu, and Crocker, 2021). The improvement from participant-specific κ, concentrated in longer utterances, suggests that the balance between evaluation of the complete utterance and local competition varies across speakers.

The present model specifies a distribution over completed utterances and evaluates the full response space. A process account must additionally explain how speakers generate candidate utterances and allocate computation to evaluating them. Manipulating planning time, and recording speech onset and eye movements, could test whether information in later adjectives affects which adjective speakers produce first.

6.2 Semantic Stability in Reference Production

The context-fixed models predicted production more accurately than models that recomputed the size comparison class during composition, as proposed by the compositional account. For the participant-specific models, this advantage chiefly concerned adjective content and utterance length; the difference in order conditional on content was small and uncertain. The result makes the maintenance of a scene-based size comparison a plausible component of reference production in this task. A speaker could establish which objects count as large or small during scene apprehension and retain that comparison while selecting adjectives.

The full scene and marked target provide information from which this comparison can be formed before production begins. In discourse, a subset of referents may become relevant through a preceding utterance, potentially changing the context against which size is evaluated. Varying when contextual information becomes available would help identify when speakers retain an initial comparison and when they revise it. Independently varying lexical subjectivity and visual discriminability would further distinguish uncertainty about a property's interpretation from difficulty perceiving it in a particular scene.

6.3 Available Alternatives and Communicative Efficiency

The simulation returns to the case that motivated the study, in which *blue* alone identifies the target but speakers who mention both properties still say *big blue*. The simulation shows that the availability of shorter utterances changes which of the same two adjective orders communication favours. The value of an order depends on which other utterances the speaker can produce and which alternatives the listener expects the speaker to have considered. This dependence gives a computational basis for studying ordering within the full distribution of content and length.

An account of reference production must consequently explain how alternatives become available. An utterance-level plan could direct consideration towards properties relevant to the referential goal, while local competition

influences which property is mentioned next.

The present data come from German, in which adjectives precede the noun. In languages with postnominal adjectives, the noun-adjacent adjective is also the first adjective the listener hears. Subjectivity-based ordering preferences have been reported for the mirror-image order in such a language (Trainin and Shetreet, 2021). Incremental pragmatics, by contrast, still favours placing the more discriminating adjective first. The two accounts therefore agree when the noun-adjacent property is also the more discriminating one, and conflict when it is not. A postnominal version of the simulation, in which composition follows the surface order, shows that updating then weakens rather than strengthens the listener-based size-first preference (Appendix G.2). Word order also affects which properties speakers mention. Rubio-Fernández, Mollica, and Jara-Ettinger (2021) found that English speakers produced more redundant colour adjectives than Spanish speakers, for whom colour follows the noun. Waldon and Degen (2021) captured this difference with an incremental RSA production model. Listeners of prenominal and postnominal languages also integrate visual context in different orders (Rubio-Fernandez and Jara-Ettinger, 2020). Postnominal languages provide a test of plan-guided production in which the scenes that create a conflict between the two accounts differ from those in German. Freer production with new participants and materials would test how the plan-guided account extends beyond these German prenominal utterances.

The broader contribution is an account of adjective ordering embedded in the construction of a useful referential utterance. In the model, scene-based meanings, stable order expectations, and local adjective competition determine which properties are mentioned and how they are arranged. The plan-guided framework connects these choices to the evaluation of a complete utterance, while the simulation shows how available alternatives affect the interpretation of an utterance. Together, these findings motivate an account of boundedly rational communication in which partial planning supports good-enough production.

7 Data Availability Statement

Data, analysis scripts, model code, posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the production-model inputs and outputs, documentation of preprocessing and input reconstruction, and the scripts required to reproduce the reported behavioural and predictive analyses.

8 Author Note

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A Supplementary Experimental Methods and Results

A.1 Participants

All studies were conducted online via Prolific, with compensation set slightly above the platform's median level. Participants reported German as their native language.

Original slider study. The original slider data are those reported by Schlotterbeck and Wang (2023). The retained sample included 70 participants in the low- and 64 in the high-discriminability group. Exclusions addressed missing records, extreme completion times, and failures on control or task-relevance checks (Table 1).

Slider replication. The replication used the same task, materials, and exclusion criteria with an independent participant sample.

Production experiment. The retained production sample comprised 60 participants with low size discriminability and 53 with high size discriminability. Participants from either slider study were ineligible. Exclusions addressed extreme completion times, missing or unannotatable responses, and more than 80% identical responses.

Table 1 summarises the participant exclusions and the observations included in each analysis.

Table 1: Participant exclusions and observations included in each analysis.

Experiment	Starting and final N	Participant exclusions	Observations included in the manuscript
Original slider study	$150 \rightarrow 134$	More than three missing trial-list entries; extreme completion time; more than three control-item failures; more than three "none fits" responses on experimental items.	All retained critical trials: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Slider replication	$132 \rightarrow 104$	Same criteria as the original study.	All retained replication trials: 8,379 critical observations. Reported size–colour slider subset: 2,786 observations.
Production experiment	$149 \rightarrow 113$	Extreme completion time; more than 80% identical responses; more than 10% missing or unannotatable responses.	Full production dataset: 9,100 annotated observations. Reported size–colour behavioural subset: 3,034 observations.

A.2 Materials

The noun remained *Aufkleber*, and size, colour, and form varied across counterbalanced lists. Critical scenes were realised through systematic permutations of colour and form values applied to 27 six-object scene templates. This varied the surface displays while preserving the referential structure of each condition.

The slider stimulus inventory contains six lists of 180 main trials: 81 critical trials and 99 fillers and controls per list, plus three practice displays. Lists 1–3 use high size discriminability and lists 4–6 use low size discriminability. Each list contains nine critical trials for each combination of the three adjective families and three referential conditions. The large-size band comprises $\{9, 10\}$ in both groups; the smaller-size band is $\{1, 2, 3\}$ at high discriminability and $\{1, 2, 3, 4, 5, 6\}$ at low discriminability. Thus, contrasts between the large and smaller bands can span 6–9 recorded units in the high-discriminability materials and 3–9 units in the low-discriminability materials.

The slider-rating materials used the fixed adjective vocabulary in Table 2.

Table 2: Adjective vocabulary in the slider-rating item descriptions.

Property class	German adjective forms displayed in the item text
Size	*großen*, *kleinen*
Colour	*blauen*, *orangen*, *grauen*, *braunen*, *grünen*, *schwarzen*
Form	*runden*, *diamantförmigen*, *herzförmigen*, *quadratischen*, *sternförmigen*, *dreieckigen*

A.3 Design and Hypotheses

The studies shared the reference-game setting and display design described in Section 2. For slider judgements, greater referential usefulness of size predicts stronger size-first preference, yielding the ordering size-sufficient $>$ both-necessary $>$ colour-sufficient; a stable size-first preference predicts ratings above neutrality even when colour alone identifies the target. The slider interaction contrasts assess how discriminability effects differ across contexts, with the replication testing whether the pattern in the original data recurs.

A.4 Procedure

Each study began with three practice trials with feedback. Main trials were divided into four blocks of approximately equal length, with self-paced breaks and independently randomised trial order within each block. In the original slider study, participants completed 81 critical trials and 99 filler or control trials (180 total), with a mean completion time of 27.6 minutes. Control trials had an unambiguous expected response; the additional "none fits" option allowed participants to indicate that neither utterance identified the target. Frequent use of this option on experimental trials was an exclusion criterion, as detailed in Table 1. The replication used the same 1,080 combinations of list, item, and condition.

On each critical trial, two phrases differing in adjective order appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left–right presentation was randomised across trials. Ratings were centred so that zero indicated neutrality, positive values favoured size-first, and negative values favoured colour-first order, on a scale from -0.5 to 0.5.

Participants typed adjective material into the frame *den ___ Aufkleber*. Practice trials introduced the relevant colour and form vocabulary. In the production experiment, participants completed 81 critical and 54 filler or control trials (135 total), with a mean completion time of 32.4 minutes.

A.5 Data Analysis

Response coding. Production responses were annotated using the 15 ordered adjective categories listed in Table 3. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size utterances, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table 3: Production annotation scheme. In the data files, `D` corresponds to the size adjective S; `C` and `F` correspond to colour and form.

Length	Codes
One adjective	`D`, `C`, `F`
Two adjectives	`DC`, `DF`, `CD`, `CF`, `FD`, `FC`
Three adjectives	`DCF`, `DFC`, `CDF`, `CFD`, `FDC`, `FCD`

The categorical response variable for the computational comparison is the ordered utterance label $y_i \in \{1, \ldots, 15\}$. For each binary behavioural outcome, matching responses were coded as 1 and all other annotated size–colour responses as 0.

Behavioural models. The behavioural hypotheses were evaluated with Bayesian hierarchical models. All behavioural analyses used the size–colour subsets: 3,485 observations in the original slider study, 2,786 in the replication, and 3,034 in the production experiment. For the slider analyses, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. Let $\mathbf{x}_{ij}$ encode referential context, size discriminability, and their interaction, and let $\mathbf{z}_{ij}$ encode the participant intercept and context slopes. The slider models used a Gaussian likelihood with

$$y_{ij} \sim \mathrm{Normal}(\mu_{ij}, \sigma), \qquad \mu_{ij} = \mathbf{x}_{ij}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ij}^{\mathsf{T}}\mathbf{b}_i, \qquad \mathbf{b}_i \sim \mathrm{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{participant}}).$$

For production, size-initiality indicated whether the coded response began with the size adjective. Redundant adjective use indicated a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts. Both binary outcomes used a Bernoulli likelihood with a logit link and an item intercept c_k:

$$y_{ijk} \sim \mathrm{Bernoulli}(\pi_{ijk}), \qquad \mathrm{logit}(\pi_{ijk}) = \mathbf{x}_{ijk}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ijk}^{\mathsf{T}}\mathbf{b}_i + c_k, \qquad c_k \sim \mathrm{Normal}(0, \sigma_{\mathrm{item}}).$$

For the slider studies, the interaction was represented by two posterior contrasts: the high-minus-low discriminability effect in the both-necessary context minus the corresponding effect in the size-sufficient context, and the high-minus-low effect in the colour-sufficient context minus the effect in the size-sufficient context. The production interactions were planned contrasts on the posterior-predictive probability scale. For size-initiality, the contrast was the high-minus-low discriminability effect in the both-necessary context minus the mean effect in the two one-property-sufficient contexts. For redundant adjective use, the contrast was the low-minus-high discriminability effect in the size-sufficient context minus the mean effect in the both-necessary and colour-sufficient contexts.

The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, fixed-effect priors $\mathrm{Normal}(0, .30)$, $\mathrm{HalfNormal}(.20)$ priors on participant standard deviations, and a $\mathrm{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\mathrm{Normal}(0, 2)$ intercept prior, $\mathrm{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\mathrm{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\mathrm{LKJ}(2)$ prior. Four chains run in parallel each provided $1{,}000$ retained draws after $1{,}000$ warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\widehat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned contrasts were then evaluated on the rating or response-probability scale.

A.6 Detailed behavioural results

Tables 4 and 5 report posterior medians and 95% credible intervals for the behavioural contrasts. Slider estimates are in centred-rating units; production estimates are differences in response probability. The column P reports the posterior probability of the direction stated in each row; a dash indicates that a directional probability was not separately reported for that contrast in the main analysis summary. Interaction estimates compare effects across contexts, whereas within-context contrasts compare the two discriminability groups in one context.

A.6.1 Slider ratings and replication

Both slider studies supported the predicted referential-context gradient, including a residual size-first preference in colour-sufficient displays (Figure 7). The original data also showed a credible interaction in the comparison of size-sufficient and both-necessary displays. Higher discriminability was associated with a shift towards size-first ratings in size-sufficient displays and towards colour-first ratings in both-necessary displays, although the credible intervals for both within-context effects included zero. The interaction therefore concerned the difference between these effects; each individual effect was less precisely estimated. The corresponding interaction between colour-sufficient and size-sufficient displays was less conclusive.

The replication did not clearly reproduce either interaction contrast. Its estimated effect of higher size discriminability was positive in all three contexts, with intervals above zero in size-sufficient and colour-sufficient displays and an interval spanning zero in both-necessary displays. Thus, clearer size differences could strengthen size-first preference in this replication, with uncertainty about variation in that effect across contexts. The two studies used the same Bayesian hierarchical model specification. Estimates in Table 4 are posterior medians, with contrasts expressed in centred-rating units; directional probabilities refer to the stated contrast under the fitted model.

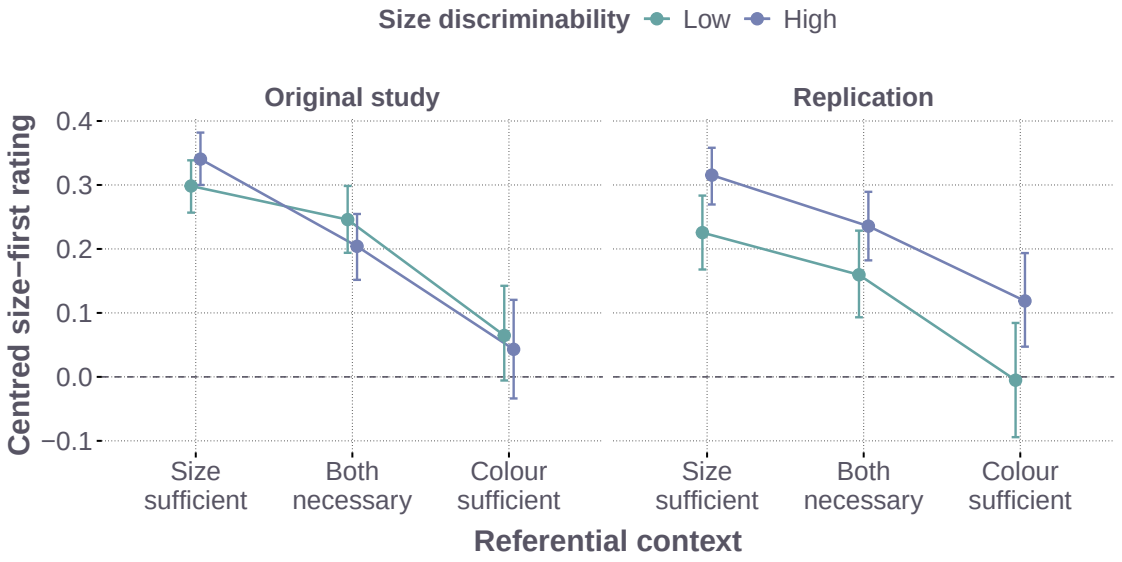


Figure 7: Ordering judgements in the original slider study and its replication by referential context and size discriminability. Points show posterior medians of expected centred ratings for a new participant; error bars show 95% credible intervals. Positive values indicate size-first preference; zero indicates neutrality. The two studies use the same model specification and interval definition.

Table 4: Slider-rating contrasts, with interactions followed by context contrasts and within-context discriminability effects. Discriminability effects are high minus low. Interaction rows subtract the size-sufficient discriminability effect from the effect in the named context.

Study	Contrast	Median	95% CrI	P (direction)
Original	Interaction: both-necessary	$-.085$	$[-.151, -.022]$	.995 (< 0)
Original	Interaction: colour-sufficient	$-.063$	$[-.172, .039]$	.885 (< 0)
Original	Size-sufficient minus both-necessary	.094	$[.061, .127]$	—
Original	Both-necessary minus colour-sufficient	.171	$[.131, .208]$	—
Original	Colour-sufficient rating relative to zero	.054	$[.003, .109]$	.982 (> 0)
Original	Discriminability: size-sufficient	.042	$[-.013, .099]$	—
Original	Discriminability: both-necessary	$-.043$	$[-.118, .028]$	—
Original	Discriminability: colour-sufficient	$-.021$	$[-.129, .082]$	—
Replication	Interaction: both-necessary	$-.014$	$[-.097, .071]$	.623 (< 0)
Replication	Interaction: colour-sufficient	.034	$[-.082, .157]$	.285 (< 0)
Replication	Size-sufficient minus both-necessary	.073	$[.029, .115]$	—
Replication	Both-necessary minus colour-sufficient	.140	$[.090, .190]$	—
Replication	Colour-sufficient rating relative to zero	.057	$[.001, .116]$	.977 (> 0)
Replication	Discriminability: size-sufficient	.090	$[.014, .160]$	—
Replication	Discriminability: both-necessary	.076	$[-.011, .161]$	—
Replication	Discriminability: colour-sufficient	.124	$[.016, .240]$	—

The posterior probability of the graded context order was greater than .999 in the original study and .999 in the replication. Observed mean centred ratings in the replication were .272 in size-sufficient, .197 in both-necessary, and .054 in colour-sufficient contexts. Figure 8 shows the observed ratings in the original study; Figure 7 shows fitted cell expectations with 95% credible intervals for both slider studies.

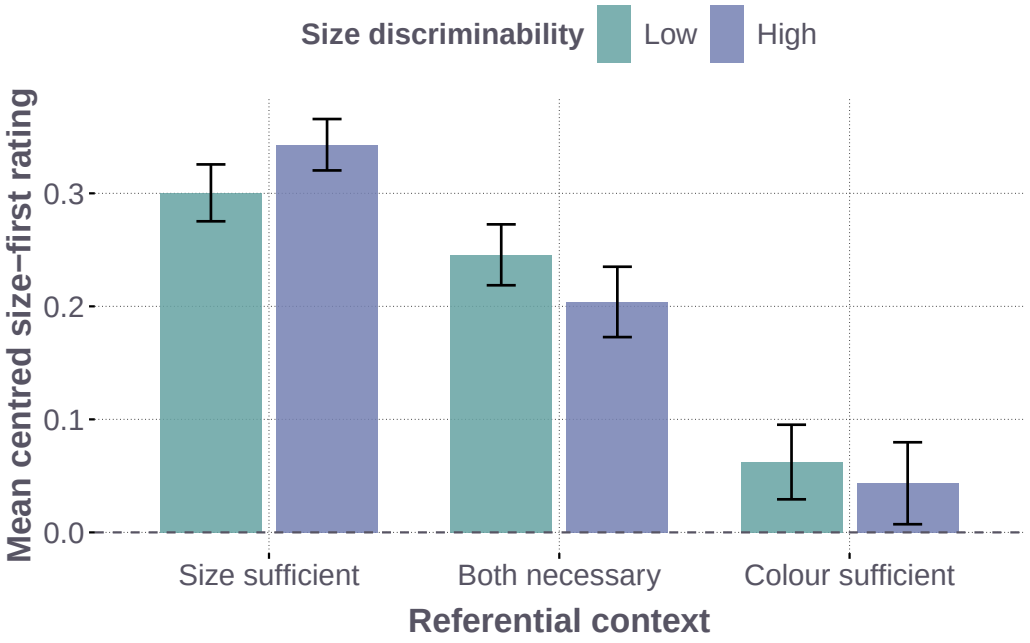


Figure 8: Observed ratings in the original slider study. Mean centred trial ratings by referential context and size discriminability, with descriptive 95% t intervals; positive values indicate size-first preference.

A.6.2 Production contrasts

The planned redundancy interaction showed a larger increase under low discriminability in size-sufficient displays than in the other two contexts (Figure 3, Panel B; Table 5). Within size-sufficient displays, lower discriminability increased redundant adjective use; the corresponding increases in both-necessary and colour-sufficient displays were smaller, and their credible intervals included zero.

For size-initiality, the high-minus-low discriminability effect was larger when both properties were necessary than when one property was sufficient (Figure 3, Panel C). Higher discriminability increased size-initial responses in both-necessary displays. The effect in size-sufficient displays was close to zero and uncertain, whereas colour-sufficient displays showed a small decrease. Averaged across discriminability conditions, size-initial responses were more probable in size-sufficient than in both-necessary displays, and more probable in both-necessary than in colour-sufficient displays. Size-initiality includes utterances consisting of size alone. Among responses containing both size and colour, size-first utterances remained more frequent, locating context sensitivity in adjective selection, length, and initial position alongside this conditional order preference.

Observed redundancy proportions in size-sufficient displays were .90 under low and .63 under high size discriminability. Observed size-initial proportions in both-necessary displays were .49 under low and .65 under high discriminability. These descriptive proportions differ from the posterior-predictive contrasts in Table 5, which integrate over participant and item variation.

A descriptive check of the complete production data also showed group differences in the colour–form family. In form-sufficient displays, form-initial responses were more frequent in the high- than in the low-discriminability group (.77 vs. .71). Because discriminability was manipulated between participants, these differences can also reflect group variation in how the task was approached. The production model represents participant variation alongside the visual manipulation.

Table 5: Production contrasts on the response-probability scale. The redundancy interaction compares the low-minus-high difference in size-sufficient displays with its mean in the other two contexts. The size-initiality interaction compares the high-minus-low difference in both-necessary displays with its mean in the two one-property-sufficient contexts.

Contrast	Median	95% CrI	P (direction)
Redundancy: planned interaction	.153	[.072, .236]	$> .999$ (> 0)
Redundancy: size-sufficient, low minus high	.234	[.151, .318]	$> .999$ (> 0)
Size-initiality: planned interaction	.169	[.077, .256]	$> .999$ (> 0)
Size-initiality: both-necessary, high minus low	.141	[.036, .242]	.997 (> 0)
Size-initiality: size-sufficient, high minus low	$-.021$	[$-.101$, .057]	—
Size-initiality: colour-sufficient, high minus low	$-.032$	[$-.068$, $-.006$]	—
Size-initiality: size-sufficient minus both-necessary	.193	[.139, .248]	—
Size-initiality: both-necessary minus colour-sufficient	.531	[.459, .601]	—

B Production Model Specification

B.1 Representation of Experimental Displays

The model represents each recorded display as a 6×3 array of size, colour, and form values, with the intended referent indexed first.

B.2 Graded Size Meaning

Size is represented by a graded threshold over the comparison class $\mathcal{C}$ (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019):

$$\theta_\delta(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta\left(Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C})\right), \tag{8}$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. For an object of size s_o, graded membership is

$$m_{\mathrm{S}}(o \mid \mathcal{C}) = \Phi\left(\frac{s_o - \theta_\delta(\mathcal{C})}{w\sqrt{s_o^2 + \theta_\delta(\mathcal{C})^2 + \epsilon}}\right), \qquad w = .6856, \quad \epsilon = 10^{-8}, \tag{9}$$

where Φ is the standard normal cumulative distribution function. The weighted quantiles are the first observed sizes at which cumulative comparison-class probability reaches .20 and .80. The same width w is used in both discriminability groups; the recorded sizes enter this function directly. For colour and form, matching objects receive membership ν and nonmatching objects $1 - \nu$, with $\nu_{\mathrm{C}} = .59$ and $\nu_{\mathrm{F}} = .50$. Each prefix is interpreted from the uniform object prior, applying its modifiers once in noun-outward order. Context updating changes the comparison-class weights used for the size quantiles as inner modifiers are applied. Equations (2), (3), (5) and (6) define how this meaning enters the literal interpretation L and the three production accounts.

Figure 9 separates fixed theoretical assumptions, estimated parameters, participant variation, and observed responses. The fixed variables A and R index production architecture and semantic regime. For participant i on trial t, the model maps the display and target X_{it} and the stable-order score $\Omega(u)$ to a probability $\pi_{it}(u)$ over the fifteen possible utterances, from which the observed response y_{it} is drawn. The shared coefficients govern visual salience, adjective selection, and utterance length.

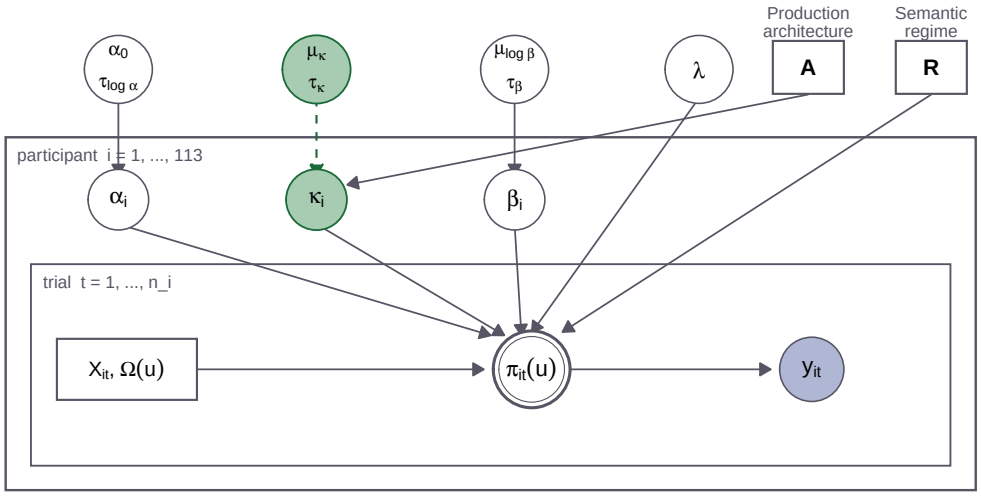


Figure 9: Hierarchical structure of the production model and the restrictions used in the matched comparisons. Solid circles represent estimated quantities, squares represent fixed model or trial inputs, the double circle represents the deterministic utterance distribution, and the shaded circle represents the observed response. The dashed κ hierarchy applies to the plan-guided model with participant-specific κ_i.

B.3 Component Inventory

Table 6: Components of the production models.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Graded support for matching colour
Form reliability	Fixed neutral	Form production supported by the length and selection terms
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Length and selection coefficients	Estimated	Support for sufficient single adjectives, form use, and length and discriminability effects
κ or κ_i	Estimated in plan-guided production	Weight of local adjective competition, shared or participant-specific

The form semantics, stable-order feature, and length and selection terms were specified on the present data and are identical across all compared production architectures and semantic regimes. The population pragmatic-optimality parameter has prior $\alpha_0 \sim \mathrm{HalfNormal}(3)$, and the population log stable-order weight has prior $\mu_{\log\beta} \sim \mathrm{Normal}(0, .35)$. Participant effects are noncentred: $\alpha_i = \alpha_0 \exp(\tau_{\log\alpha} z_{\alpha i})$ and $\beta_i = \exp(\mu_{\log\beta} + \tau_\beta z_{\beta i})$, with independent standard-normal z values, $\tau_{\log\alpha} \sim \mathrm{HalfNormal}(.15)$, and $\tau_\beta \sim \mathrm{HalfNormal}(.5)$. The shared plan-guided weight has prior $\kappa \sim \mathrm{Uniform}(0, 1)$. In the model with participant-specific κ_i, $\mu_\kappa \sim \mathrm{Uniform}(0, 1)$ and $\kappa_i = \mathrm{logit}^{-1}(\mathrm{logit}(\mu_\kappa) + \tau_\kappa z_{\kappa i})$, with $\tau_\kappa \sim \mathrm{HalfNormal}(.5)$ and independent $z_{\kappa i} \sim \mathrm{Normal}(0, 1)$. The salience coefficient has a $\mathrm{HalfNormal}(1)$ prior and the salience-continuation coefficient a $\mathrm{HalfNormal}(.75)$ prior. The sufficient-single-adjective and form-use coefficients each have a $\mathrm{HalfNormal}(1.5)$ prior; the three-adjective penalty and the two size-reliability coefficients each have a $\mathrm{HalfNormal}(1)$ prior. Threshold selectivity is fixed at $\delta = .50$; colour reliability, form reliability, and perceptual blur take the values specified above.

Visual salience. The salience feature combines a fixed preference across properties with their contrast in the displayed scene. For the target o^*, let s_o, c_o, and f_o be an object's recorded size, colour, and form, and let h indicate high size discriminability. The three contrast scores are

$$r_{\mathrm{S}} = \frac{1+h}{2}\,\mathrm{logistic}\!\left(\frac{s_{o^*} - \max_{o\neq o^*} s_o}{\mathrm{sd}(s) + 10^{-8}}\right), \tag{10}$$

$$r_{\mathrm{C}} = 1 - \frac{1}{5}\sum_{o\neq o^*} \mathbf{1}\{c_o = c_{o^*}\}, \qquad r_{\mathrm{F}} = 1 - \frac{1}{5}\sum_{o\neq o^*} \mathbf{1}\{f_o = f_{o^*}\}. \tag{11}$$

Here $\mathrm{sd}(s)$ is the population standard deviation across the six displayed sizes. The fixed baseline scores are $(b_{\mathrm{S}}, b_{\mathrm{C}}, b_{\mathrm{F}}) = (0, 1, .25)$. Centring across the three properties gives

$$z_a = b_a + r_a - \frac{1}{3}\sum_v (b_v + r_v).$$

A property receives greater salience when it contrasts with the distractors; the size contrast is additionally scaled by discriminability. The same feature enters prefix utility through $\lambda_{\mathrm{salience}} z_a$ and the continuation cost in the length and selection terms below.

Stable order expectations. We derived the fixed order feature from `dbmdz/german-gpt2`. For each pairing of six colour and six form adjectives, the scoring script generated all fifteen non-empty orders using those adjectives and *großer*, followed by the noun *Aufkleber*, without a surrounding sentence. It evaluated the resulting 540 strings, including repeated shorter strings, using mean token surprisal in bits. We averaged this surprisal within each order category and normalised $2^{-\bar{s}(u)}$ across the fifteen categories to obtain the fixed weights $q(u)$. For $\mathcal{U}_{A(u)}$, the set of orders expressing the same adjective set as u, the model uses

$$\Omega(u) = \log q(u) - \frac{1}{|\mathcal{U}_{A(u)}|}\sum_{v\in\mathcal{U}_{A(u)}} \log q(v). \tag{12}$$

This centring preserves relative order preferences within an adjective set and gives every single-adjective response a score of zero. The scoring script, archived token-surprisal table, aggregation code, and fixed scores and residuals are supplied with the model code. Participant-specific β_i determines the influence of these stable expectations when the model normalises across the complete response space.

Length and selection terms. The length and selection terms $T(u, X)$ connect the referential design to choices about which properties to mention and how far to continue an utterance. Let $n = |u|$ and $A(u)$ denote the length and unordered adjective set of u. For this term, X includes the display's condition labels: $d \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}, \varnothing\}$ is the property that identifies the target on its own, with $d = \varnothing$ in both-necessary conditions; c indicates a colour-sufficient condition; and $h = 1$ indicates high size discriminability ($h = 0$ for low). Using $\mathbf{1}\{\cdot\}$ for an indicator, the production models use

$$\begin{aligned} T(u, X) = \; & -\rho_{\mathrm{cont}} \sum_{t=1}^{n-1} \max(z_{w_t}, 0) + \lambda_{\mathrm{suff}} \mathbf{1}\{d \neq \varnothing,\ u = (d)\} \\ & + \lambda_{\mathrm{F}}(1-c)\mathbf{1}\{\mathrm{F} \in A(u)\} - \lambda_3 \mathbf{1}\{n = 3\} \\ & + \lambda_{\mathrm{S}} h\, \mathbf{1}\{d = \mathrm{S},\ u = (\mathrm{S})\} \\ & + \lambda_{\mathrm{SF}}(1-2h)\mathbf{1}\{d = \mathrm{S},\ A(u) = \{\mathrm{S}, \mathrm{F}\}\}. \end{aligned} \tag{13}$$

The continuation cost sums the positive salience scores of adjectives followed by another adjective, using the same z scores as the prefix utility in Equation (3). The sufficiency term favours a single adjective that identifies the target, while the form term increases the value of form-containing utterances outside colour-sufficient conditions. The three-adjective penalty offsets the value accumulated along longer utterances. In size-sufficient conditions, the final two terms additionally favour the single size adjective at high discriminability and favour size–form pairs at low discriminability while penalising them at high discriminability. The pair term applies equally to both orders. All six coefficients are nonnegative and shared across participants within each fitted model, with the priors given above. These terms enter the completed-utterance score alongside accumulated prefix utility, local competition, and the stable-order score; normalisation over all fifteen responses turns their combined values into predictions of adjective selection, length, and order.

Response likelihood. Writing $g(u)$ for accumulated prefix utility, $i(u)$ for the sum of local log probabilities, and $B(u) = \beta_i \Omega(u) + T(u, X)$, the computed score before Laplace smoothing is $(1 - \kappa_i)g(u) + \kappa_i i(u) + B(u)$. Normalising these scores yields $\widetilde{S}_{\kappa_i}(u \mid X)$, whose endpoints obey the geometric interpolation in Equation (18). A final Laplace smoothing step mixes this distribution with a uniform distribution over the fifteen responses, giving the categorical response likelihood

$$p(y_{it} = u \mid X_{it}) = (1 - .003)\widetilde{S}_{\kappa_i}(u \mid X_{it}) + \frac{.003}{15}. \tag{14}$$

The matched global and fully incremental models fix κ at zero and one, respectively; the matched plan-guided model estimates a shared κ; the model with participant-specific κ_i estimates them for each participant. All models retain participant hierarchies for pragmatic optimality and stable-order weight, with the remaining task coefficients shared across participants.

C Production Model Comparison and Diagnostics

C.1 Inference and Predictive Comparison

Each likelihood evaluation computes all fifteen completed utterances exactly, with the listener and speaker computations vectorised and compiled in JAX. All eight production fits used four No-U-Turn Sampler (NUTS) chains on a GPU and 1,000 warm-up iterations per chain. The two global models retained 4,000 draws per chain at maximum tree depth 8; the other six models retained 1,000 draws per chain at depth 6. The eight fits had zero divergences, maximum $\widehat{R} < 1.01$, minimum bulk and tail effective sample sizes above 400, minimum Bayesian fraction of missing information (BFMI) above .64, and no observation with Pareto $k > .7$.

The pointwise leave-one-out score vectors use the same trial ordering and response support. Because observations are repeated within participants, paired score contributions are first summed within participant. A participant-level Bayesian bootstrap then assigns Dirichlet weights to the 113 participant contributions over 200,000 draws. Reported uncertainty comprises 95% credible intervals and posterior probabilities for directional contrasts. The predictive target is another response from a participant represented in the dataset, with that participant's remaining trials available to estimate their participant-specific parameters.

Table 7: Bayesian-bootstrap contrasts for the matched production architectures and the plan-guided participant hierarchy.

Contrast	ΔELPD	95% credible interval	$P(\Delta > 0)$
Plan-guided, shared κ, minus average endpoints	402.677	$[311.466, 503.098]$	$> .999$
Fully incremental minus global	-291.596	$[-684.673, -1.321]$	.024
Updating minus fixed, shared-κ grid	-43.527	$[-56.005, -31.770]$	$< .001$
Participant κ_i hierarchy, context-fixed	364.742	$[248.178, 520.967]$	$> .999$
Participant κ_i hierarchy, updating	372.285	$[255.597, 526.056]$	$> .999$

Averaging over the two semantic regimes, the plan-guided model had the highest predictive accuracy of the three architectures in .994 of the Bayesian-bootstrap draws. In the context-fixed plan-guided model, the posterior mean of the shared κ was

$$\kappa = .446, \qquad 95\% \text{ credible interval} = [.407, .486]. \tag{15}$$

C.2 Architecture-Level Response Patterns

Figure 10 shows initial-adjective cells in which the largest difference among architecture predictions is at least 6.7 percentage points and traces these differences into completed utterances. Colour-initial responses contributed 328.912 ELPD to the matched plan-guided model's advantage over the average endpoint, or 81.7% of the summed pointwise contrast. Across the displayed cells, plan-guided production often lies between the global and fully incremental endpoints.

C.3 Decomposing the Semantic Predictive Contrast

The architecture-specific updating-minus-fixed effects were -45.104 ELPD for global production, -61.683 for plan-guided production, and -23.793 for fully incremental production. Each architecture-specific 95% credible

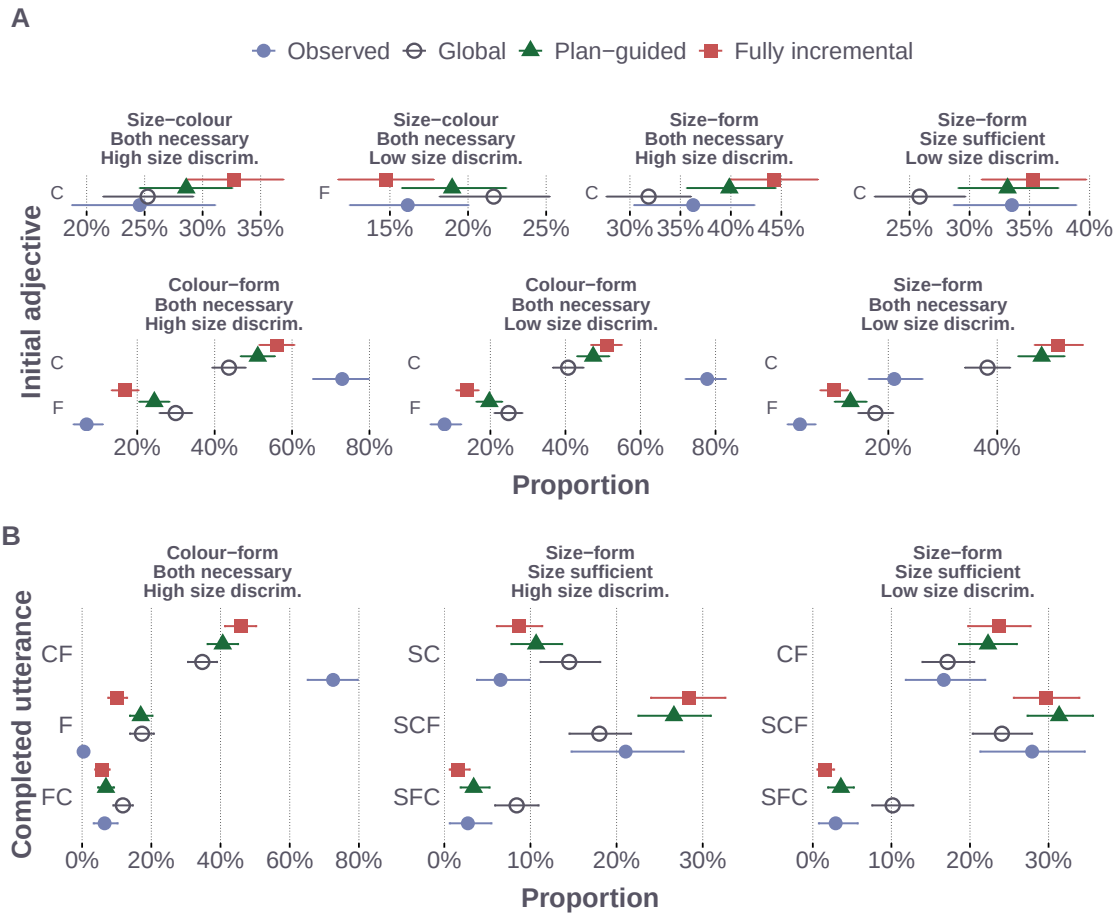


Figure 10: Qualitative comparison of the matched production architectures. Panel A shows initial-adjective cells for which the maximum difference among model predictions is at least 6.7 percentage points. Panel B shows completed utterances with differences of at least five percentage points within the three experimental conditions with the largest pairwise total-variation distance among model predictions. Response labels use S for size, C for colour, and F for form. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Horizontal scales vary across conditions.

interval excluded zero. The sequentially updating participant-specific model was lower by 54.140 ELPD, with a 95% credible interval for updating minus fixed of $[-69.042, -39.733]$.

Let $\ell(u)$ denote utterance length and $A(u)$ its unordered adjective set. For each held-out response, its predictive probability factors as

$$p_{-it}(u \mid X_{it}) = p_{-it}(\ell(u) \mid X_{it})\, p_{-it}(A(u) \mid \ell(u), X_{it})\, p_{-it}(u \mid A(u), X_{it}). \tag{16}$$

We first integrate the fifteen-response distribution using the PSIS weights for leaving out that entire response, then obtain the length and adjective-set probabilities by summation. Taking logarithms yields three additive contributions to the original response score. Paired differences are summed within participant, with 95% intervals from 200,000 Dirichlet-bootstrap draws. For the participant-specific κ_i pair, updating minus fixed was -21.549 ELPD for length, 95% credible interval $[-30.933, -12.443]$, and -31.068 for adjective set conditional on length, $[-39.202, -23.530]$. The conditional-order contribution was -1.524 ELPD, with an interval of $[-4.879, 1.774]$. Across all four semantic pairs, adjective set conditional on length makes a negative contribution to updating minus fixed. For the same pair, the original response contrast sums to -17.548 ELPD in size–colour trials, -51.172 in size–form trials, and 14.580 in colour–form trials.

Response residuals. For the two plan-guided models with participant-specific κ_i, mean absolute residuals across response cells were 3.04 percentage points under context-fixed semantics and 3.08 under updating. The largest between-model differences occurred in size-sufficient displays, in line with the held-out partition, which located the largest net disadvantage of updating in the size–form family.

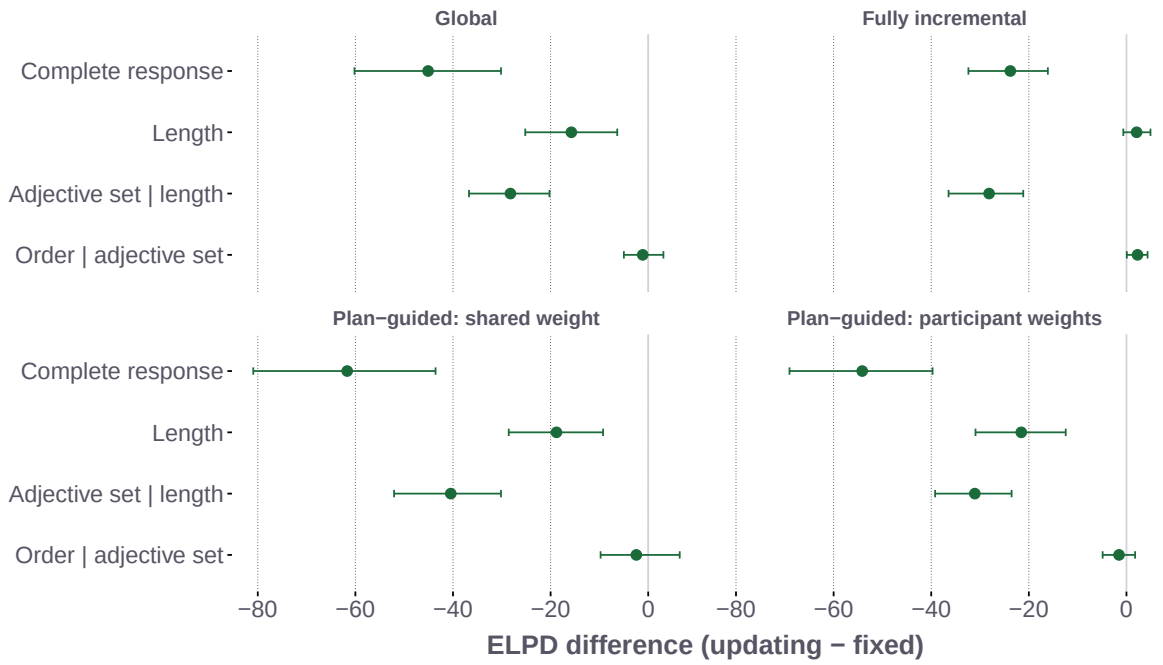


Figure 11: Additive decomposition of updating-minus-fixed response ELPD for each matched semantic pair. Length, adjective set conditional on length, and order conditional on the adjective set sum to the complete-response contrast. Points show paired score differences and intervals show participant-level Bayesian-bootstrap 95% credible intervals. The final panel uses participant-specific κ_i; the other plan-guided panel uses a shared κ.

C.4 Posterior-Predictive Adequacy

The condition-wise response distributions appear in Figure 12. Across the 270 cells formed by the fifteen completed utterances, three adjective families, three referential contexts, and two size-discriminability conditions, the posterior predictions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points and a root mean squared error of 6.0 percentage points.

In size–colour trials, the model reproduced the predominance of colour-only responses when colour was sufficient and the greater use of size alone when size was sufficient and highly discriminable. The main remaining discrepancy concerned the balance between some two- and three-adjective utterances, with size–form responses overpredicted and size–colour–form responses underpredicted.

C.5 Robustness Checks

Five robustness checks address the participant-level structure of the architecture comparison, its distribution across adjective families, the fixed colour reliability, the condition-specific length and selection terms, and generalisation to new participants. Unless stated otherwise, contrasts are context-fixed, summed within participant, and summarised with the participant-level Bayesian bootstrap described above.

Endpoint mixture benchmark. For each participant, the benchmark takes the higher held-out score of the global and the fully incremental model. Because this choice is made after seeing the scores, the benchmark favours an account in which participants differ only in which endpoint they resemble. The plan-guided model with participant-specific κ_i predicted better than this benchmark, whereas the plan-guided model with a shared κ did not (Table 8). The participant-specific model exceeded both endpoint models for 59 of the 113 participants.

Adjective families. The plan-guided advantage over the average of the two endpoint models was credible in each adjective family. Within size–colour trials, it was larger for responses that contained a form adjective than for responses that did not, but it was credible in both.

Colour reliability. The plan-guided models with context-fixed and updating semantics were refitted with colour reliability $\nu_C = .75$ instead of .59. Both models predicted worse than at $\nu_C = .59$ ($\Delta\text{ELPD} = -263.0$, 95% CrI

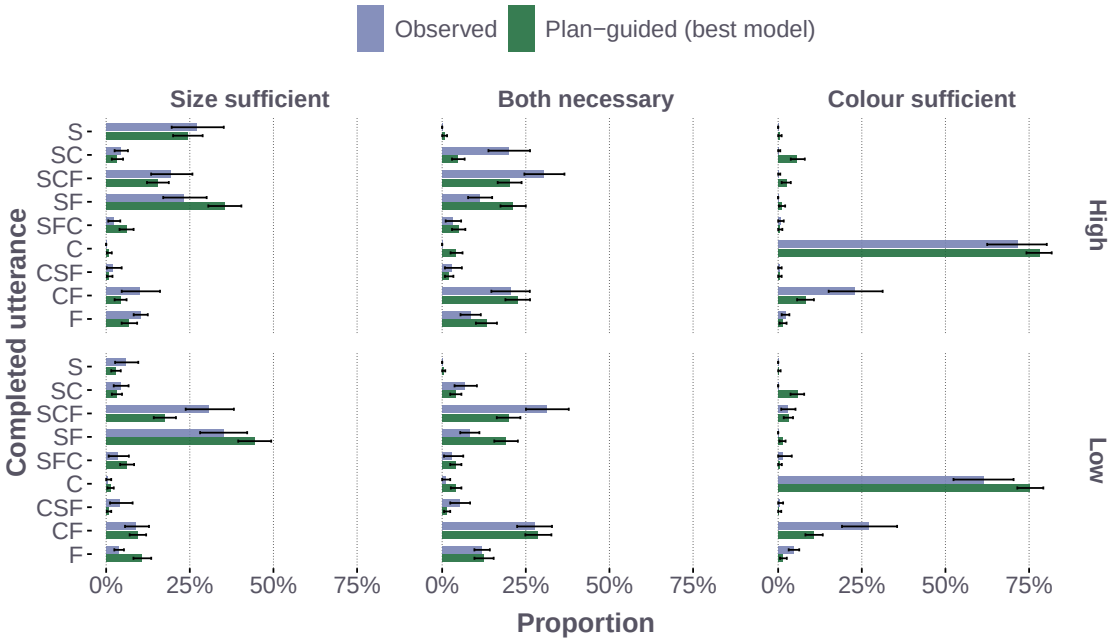


Figure 12: Observed and posterior-predicted proportions for frequent utterances in size–colour trials under the best-predicting plan-guided model. The figure includes utterances whose observed or mean posterior-predicted proportion reaches 5% in at least one context-by-discriminability condition, with proportions pooled across trials within each condition. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Adjective labels preserve surface order: S denotes size, C colour, and F form.

$[-365.7, -163.4]$ for context-fixed semantics), and updating still predicted worse than context-fixed semantics. The shared κ remained between the endpoints (posterior mean .29).

Condition-specific terms. The global, plan-guided, and fully incremental models were refitted under both semantic regimes with the two condition-specific length and selection terms fixed at zero. These models predicted worse than their counterparts with the two terms, but plan-guided production still predicted better than both endpoints and context-fixed semantics still predicted better than updating. The shared κ remained between the endpoints (posterior mean .50, 95% CrI $[.46, .55]$ under context-fixed semantics).

New participants. Participants were assigned to ten folds, stratified by size-discriminability group. Each model was refitted without the participants of one fold, and every held-out participant was scored as a new speaker. The score integrates over participant effects drawn from the fitted population distribution and is computed jointly over all of that participant's responses, using 400 posterior draws and 100 participant-effect draws per posterior draw. All 50 fits had no divergences and a maximum $\widehat{R}$ of 1.011; halving the number of participant-effect draws changed fold totals by less than 0.5 ELPD.

Table 8: Robustness checks for the production model comparison. Differences are in ELPD, summed over participants, with 95% credible intervals from the participant-level Bayesian bootstrap. Rows use context-fixed semantics unless stated otherwise.

Contrast	ΔELPD	95% CrI
Endpoint mixture benchmark		
Participant-specific κ_i minus benchmark	249.7	$[103.8, 416.1]$
Shared κ minus benchmark	-115.0	$[-286.4, 24.7]$
Participant-specific κ_i minus benchmark, updating	249.3	$[106.9, 412.7]$
Plan-guided minus average endpoint, by adjective family		
Size–colour	202.4	$[169.3, 237.2]$
responses without form	29.6	$[9.1, 53.0]$
responses with form	172.8	$[138.9, 209.1]$
Size–form	150.7	$[108.5, 193.7]$
Colour–form	63.2	$[9.0, 115.6]$
Updating minus fixed, plan-guided, by colour reliability		
$\nu_{\mathrm{C}} = .59$	-61.7	$[-80.9, -43.6]$
$\nu_{\mathrm{C}} = .75$	-45.9	$[-59.9, -32.0]$
Without condition-specific length and selection terms		
Plan-guided minus global, averaged over regimes	286.1	$[90.5, 444.6]$
Plan-guided minus fully incremental, averaged over regimes	488.6	$[308.6, 710.3]$
Updating minus fixed, plan-guided	-71.4	$[-94.5, -49.7]$
New participants (ten-fold cross-validation)		
Plan-guided minus global	271.6	$[79.5, 425.1]$
Plan-guided minus fully incremental	457.8	$[264.8, 694.3]$
Participant-specific κ_i minus shared κ	318.8	$[213.4, 455.2]$
Updating minus fixed, plan-guided	-53.8	$[-72.6, -34.8]$

D Participant Variation and Predictive Contributions

The joint model estimates pragmatic optimality, κ_i, and stable-order weight for each participant. Table 7 reports its predictive gain relative to the otherwise matched shared-κ model under each semantic regime. In the best-predicting context-fixed model, the population location of the participant-specific κ_i was $\mu_\kappa = .472$, with a 95% credible interval of $[.337, .618]$. The participant distribution was broad, with an estimated logit-scale standard deviation of 2.720, 95% credible interval $[2.277, 3.207]$.

Participants whose posterior mean κ_i lay farther from the shared $\kappa = .446$ gained more participant-level ELPD from the hierarchy. The observed correlation was $r = .618$, and the participant-level Bayesian-bootstrap mean was .639, with a 95% credible interval of $[.525, .763]$ and posterior probability above zero greater than .999. At the response level, the hierarchy gained 229.372 ELPD on two-adjective utterances, 95% credible interval $[163.012, 319.334]$, and 183.796 ELPD on three-adjective utterances, 95% credible interval $[100.757, 304.826]$. Its contribution to one-adjective utterances was -48.426 ELPD (95% credible interval $[-70.371, -28.666]$).

E Local Normalisation and Model Endpoints

Local adjective competition. For the utilities defined in Equation (3), taking the logarithm of the product of locally normalised choices gives the accumulated utility minus the local-normalisation term. Suppressing the shared referent and scene arguments,

$$\begin{aligned}\log \prod_t \frac{\exp U_t(w_t \mid u_{<t})}{\sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t})} &= \sum_t U_t(w_t \mid u_{<t}) - \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}) \\ &= U^{\mathrm{complete}}(u) - \mathcal{N}^{\mathrm{local}}(u).\end{aligned} \tag{17}$$

Relation between the endpoints. At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The stable-order score, length and selection terms, and utterance space are identical across the three architectures.

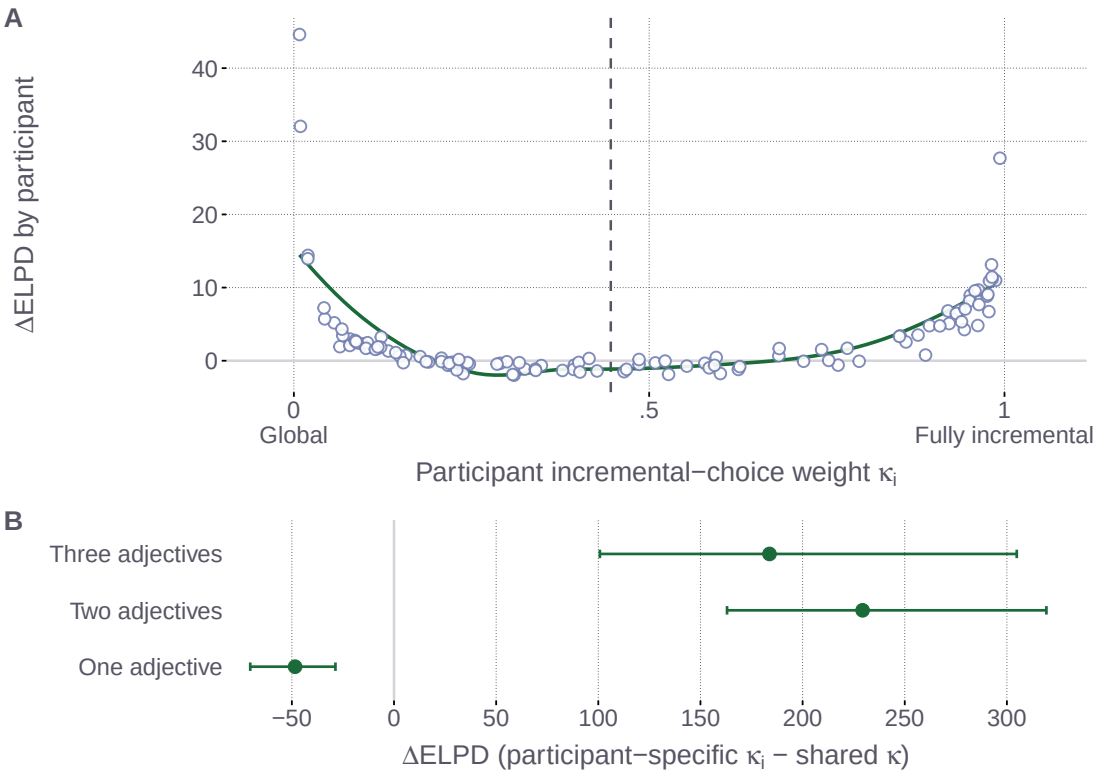


Figure 13: Where participant-specific κ_i improve prediction. Panel A relates each participant's leave-one-out gain from the κ_i hierarchy to their posterior mean κ_i; the dashed line marks the shared-κ estimate and the curve is a descriptive smooth. Panel B decomposes the total ELPD gain by observed response length. Intervals in Panel B are 95% credible intervals from the participant-level Bayesian bootstrap. Both panels use the best-predicting plan-guided model and its otherwise matched shared-κ model to localise the held-out model contrast.

Plan-guided interpolation. Before the final Laplace smoothing, the same model can be written at common parameter values as a geometric interpolation between the global and fully incremental endpoints:

$$S_\kappa(u \mid o^*, X) \propto S_{\mathrm{G}}(u \mid o^*, X)^{1-\kappa} S_{\mathrm{I}}(u \mid o^*, X)^{\kappa}, \tag{18}$$

where S_{G} and S_{I} are the endpoint distributions evaluated with the same listener, stable-order component, length and selection terms, and parameter values. The proportionality in Equation (18) renormalises the interpolated scores over the fifteen completed utterances.

F Sequential Context Updating for Size

For the hierarchical semantic path, let C_{j-1} be the context state after the syntactically inner modifiers have composed. The semantic regime determines the comparison class used for size:

$$[\![\mathrm{S}]\!]_{\mathrm{fix}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(\mathcal{C})}(o), \tag{19a}$$

$$[\![\mathrm{S}]\!]_{\mathrm{upd},C_{j-1}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(C_{j-1})}(o). \tag{19b}$$

where θ_δ derives a graded size threshold from a comparison class and $\mathcal{C}$ is the original scene, represented by the uniform prior P_0. Under updating, the literal interpretation recomputes the size threshold from the hierarchically prior state; under context-fixed semantics, it anchors the threshold to the scene prior. At each surface position, the production calculation evaluates the current prefix using this noun-outward interpretation.

G Simulation Parameter Sweeps

The original two-order simulation compares a global rule based on completed-utterance utility with an incremental rule that includes continuation-versus-stop competition. We report its scene construction, definitions, and

parameter sweeps first. Appendix G.1 then introduces the production specification on the same scenes, as in the controlled comparison in Section 5.

Each random display contains between 2 and 30 objects whose sizes are drawn from a normal distribution centred at 15.5 and truncated to $[1, 30]$. The largest object is designated as the target and assigned blue; each non-target object is independently assigned blue or non-blue with equal probability. Three levels of contextual size spread vary how distinctly the largest target differs from its competitors. The semantic grid varies colour reliability, size-threshold selectivity, and perceptual blur, which control the strength of colour-based restriction, how selectively *big* applies, and how sharply size differences are represented. In the simulation notation, k corresponds to threshold selectivity δ and w_f to perceptual blur w in Equations (8) and (9).

Figures 15 to 17 show how the three fixed semantic parameters affect communicative success across the two speaker architectures. Each figure separates the semantic regimes and averages equally over the other parameter settings. For the two-order simulation, write $q(v, o) = L(o \mid v, X)^\alpha$, with $\alpha = 1$ and equal utterance costs. For distinct adjectives a, b, denote the original global and incremental rules by S_G and S_I:

$$S_G(ab \mid o, X) = \frac{q(ab, o)}{q(ab, o) + q(ba, o)}, \tag{20}$$

$$R_I(ab \mid o, X) = \frac{q(a, o)}{q(a, o) + q(b, o)} \frac{q(ab, o)}{q(ab, o) + q(a, o)}, \tag{21}$$

$$S_I(ab \mid o, X) = \frac{R_I(ab \mid o, X)}{R_I(ab \mid o, X) + R_I(ba \mid o, X)}. \tag{22}$$

The second factor of R_I is the continuation-versus-stop choice, and the final normalisation conditions on producing a completed two-adjective order. The original global rule supplies the terminal utility used in the first row of Panel A in the main comparison. The production global model accumulates utility across adjective positions; its incremental endpoint assigns utterance length through the completed-response distribution. The pragmatic listener uses a uniform referent prior, giving $L^{\text{prag}}(o \mid ab, X) = S(ab \mid o, X) / \sum_r S(ab \mid r, X)$.

The reproducible sweep uses seed 20260906 and samples 1,000 independent scenes for each context-size and size-spread combination. Each scene is reused across semantic parameters, semantic regimes, and speakers, yielding 24,000 independent scenes and 12,000,000 model evaluations. Monte Carlo standard errors are calculated after averaging parameter settings within scenes; repeated evaluations of one scene do not count as independent samples.

Figure 14 shows how the size-first preference changes across the broader random contexts, in which colour generally restricts the comparison class without uniquely identifying the target. Wider size distributions generally make the largest target more distinguishable from its competitors; clustered sizes reduce this distinction.

The original global rule with context-fixed semantics provides a symmetry baseline: both orders have the same fixed intersective meaning and equal cost, so they yield the same target probability. Sequential context updating breaks this symmetry because *big blue* interprets size relative to the comparison class established by colour, producing a size-first preference under global production. The preference is strongest in small contexts with widely separated sizes, where restricting the comparison class has the clearest consequences for which objects count as big, and weakens as contexts grow or sizes cluster.

The original incremental rule adds a contribution from local informativity along the surface sequence. With closely clustered sizes, this rule yields a colour-first preference in pragmatic target recovery. As sizes spread apart, the colour-first preference weakens. After averaging over the semantic parameter settings in Figure 14, the original incremental rule yields colour-first preferences in some contexts; the original global rule remains symmetric under fixed semantics and yields a size-first preference under updating. Under the original incremental rule, sequential context updating increases the size-first preference in target recovery when the colour-restricted comparison class changes size interpretation sufficiently strongly. The parameter sweeps in Figures 15 to 17 show that the same competition extends across colour reliability, size-threshold selectivity, and perceptual blur.

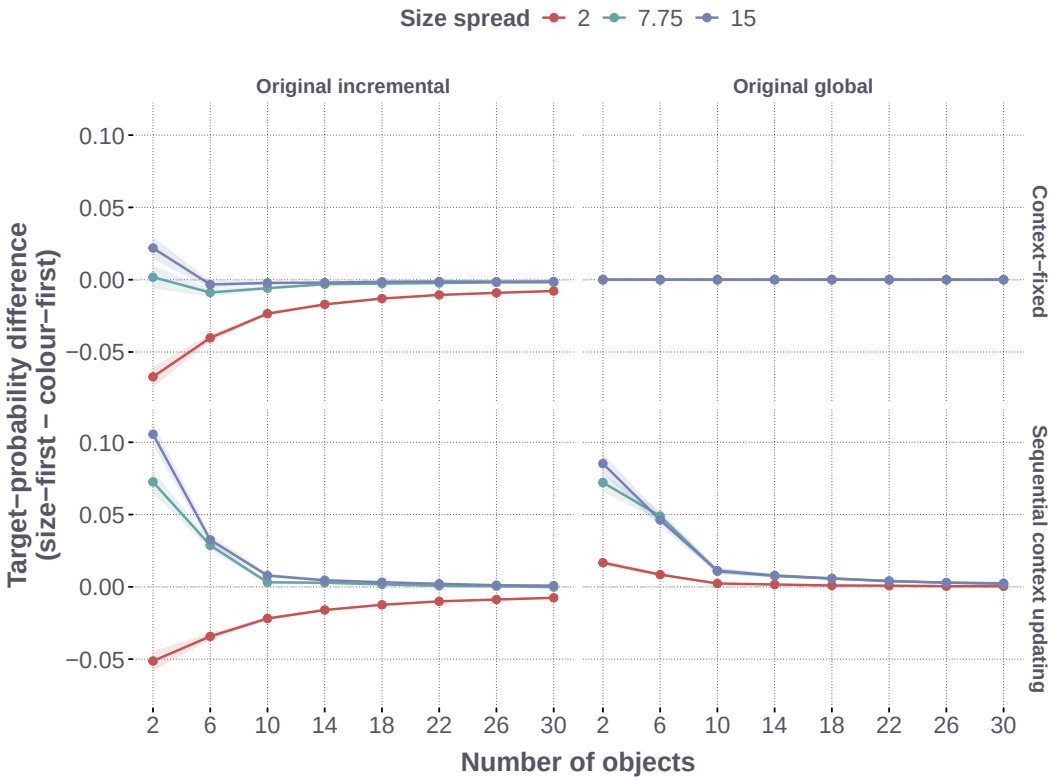


Figure 14: Size-first preference Δ_{SF} across context size and contextual spread, shown for the original two-order global and incremental rules under context-fixed semantics and sequential context updating. Wider size distributions make size a more reliable identifying cue; clustered distributions make it less reliable. Each point averages over the semantic sensitivity settings; ribbons show ± 1.96 Monte Carlo standard errors calculated after averaging those settings within each sampled scene. The complete parameter grid appears in Table 9. Positive values indicate a higher model-assigned target probability for the size-first order.

Table 9: Simulation parameter grid

Parameter	**Grid values**	**# levels**
Original two-order rule	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
Colour reliability ν	0.90, 0.92, 0.94, 0.96, 0.98	5
Threshold selectivity k	0.10, 0.30, 0.50, 0.70, 0.90	5
Perceptual blur w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Scenes per context-size/spread pair	1,000	
Total model evaluations	**12,000,000**	

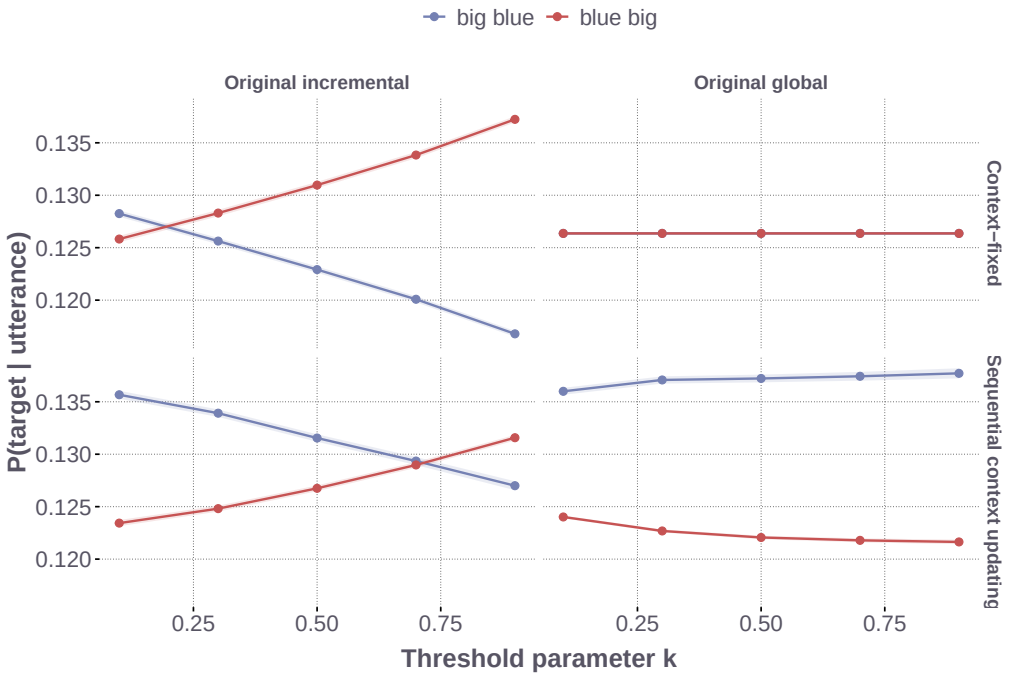


Figure 15: Effect of the threshold parameter k on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

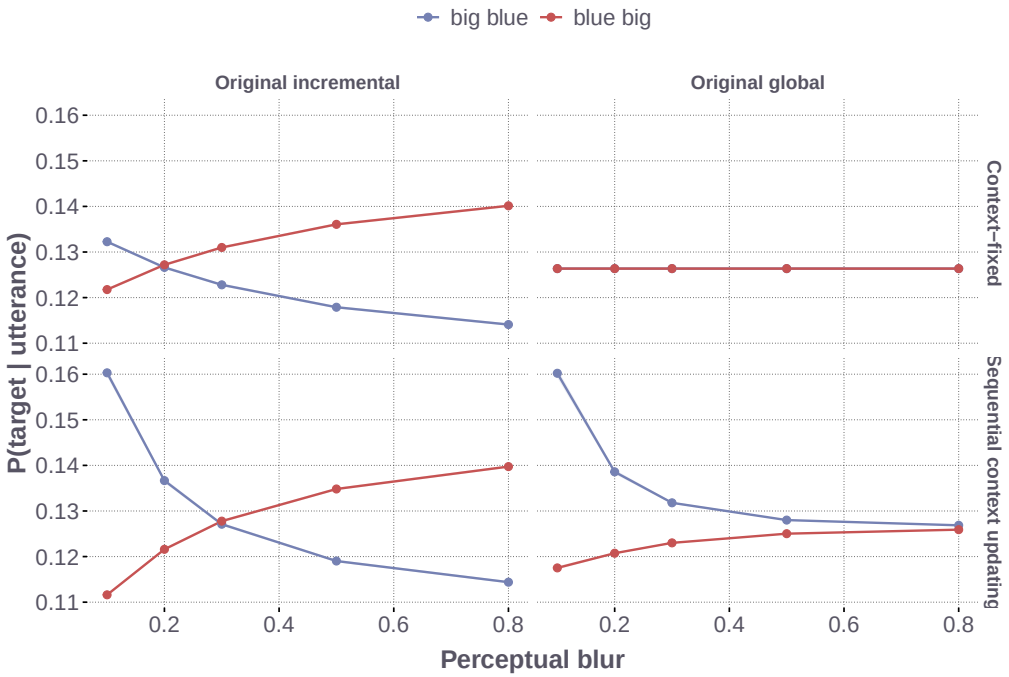


Figure 16: Effect of perceptual blur w_f on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

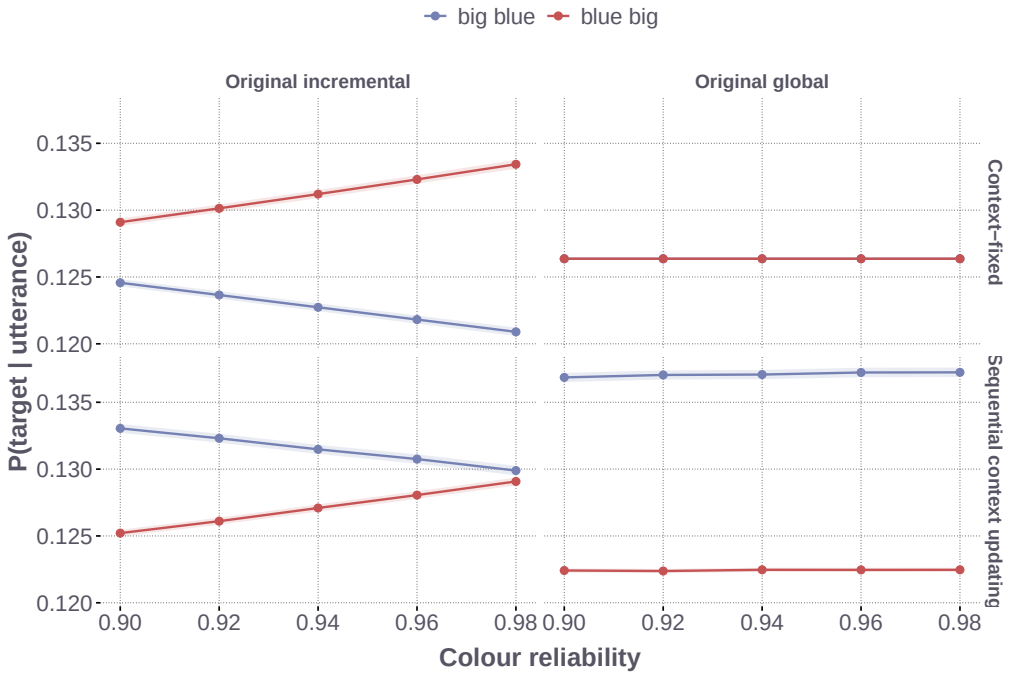


Figure 17: Effect of colour reliability ν on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

G.1 Controlled Changes on the Same Random Scenes

The terminal utility differs from the prefix utility:

$$U_{\text{term}}(u, o, X) = \alpha \log L(o \mid u, X), \tag{23}$$

$$U_{\text{prefix}}(u, o, X) = \alpha \sum_{t=1}^{|u|} \log L(o \mid u_{1:t}, X). \tag{24}$$

Both scores are globally normalised over completed utterances. The prefix score with $\kappa = 0$ is the global model used in the main text; $\kappa = .5$ and $\kappa = 1$ give the plan-guided and fully incremental simulation models. The terminal score supplies a control for the additional contribution of intermediate adjective information.

The controlled comparisons reuse all 24,000 scenes from the original sweep and the target-recovery measure in Equation (7). Each adjective retains the same extension when the intended referent changes; the referent prior is uniform. The response spaces contain two orders, four size–colour responses including the single adjectives, or the fifteen production responses. For expanded spaces, the pragmatic listener uses each order's probability under the full speaker distribution, without first conditioning on the two orders. The comparisons include the terminal utility and the prefix utility at $\kappa = 0, .5, 1$, using the original 125 semantic settings or the production constants. Prefix scores and local candidate sets are recomputed for each utterance space. They retain the production convention of no separate STOP action and final normalisation over completed paths. This isolates the computational components without assigning fitted participant effects or experimental task labels to random scenes.

The main comparison holds pragmatic optimality at $\alpha = 1$ and excludes stable-order expectations, salience, length and selection terms, and final Laplace smoothing. Form reliability is fixed at .50. Panel A of the main figure separates the effects of evaluating information at each adjective position and expanding the utterance space, while holding global production fixed. Panel B compares global, plan-guided, and fully incremental production using $\kappa = 0, .5, 1$, respectively, at the production semantic constants. The production constants are colour reliability .59, threshold selectivity .50, and perceptual blur .6856. Table 10 reports the mean preferences in both panels of Figure 6. The prefix-utility rows specify the architecture through κ; the terminal utility uses global production. For the fifteen-response global model at the production constants, the paired updating-minus-fixed difference was 0.0037 percentage points, with a Monte Carlo interval of $[0.0007, 0.0067]$.

Table 10: Mean size-first preference in pragmatic target recovery in percentage points on the same 24,000 random scenes.

Utility	Responses	Semantic settings	Context-fixed	Updating
Terminal	2	Parameter grid	0.000	1.141
Prefix ($\kappa = 0$)	2	Parameter grid	-2.718	-1.838
Prefix ($\kappa = 0$)	4	Parameter grid	2.257	2.772
Prefix ($\kappa = 0$)	15	Parameter grid	2.589	3.104
Prefix ($\kappa = 0$)	15	Production constants	1.891	1.894
Prefix ($\kappa = .5$)	15	Production constants	1.455	1.457
Prefix ($\kappa = 1$)	15	Production constants	0.960	0.962

The main figure uses the production semantic implementation, whose probability floors differ slightly from those of the original sweep in Figure 14.

Figure 18 reports the fifteen-response comparison across utility rules and local-normalisation weights. The final panel adds the same stable-order coefficient, $\exp(.962357)$, to every model at the production semantic constants. This setting is the fixed model's population log-weight location. At $\kappa = 0$, adding this term gives mean preferences of 2.154 and 2.157 percentage points for fixed and updating semantics, respectively. The positive preferences are already present without this term. All other salience, length, and selection terms remain zero, and $\alpha = 1$ throughout.

For every contrast, semantic parameter settings are averaged within each scene before calculating Monte Carlo uncertainty. The overall mean gives equal weight to the 24 context-size and size-spread groups.

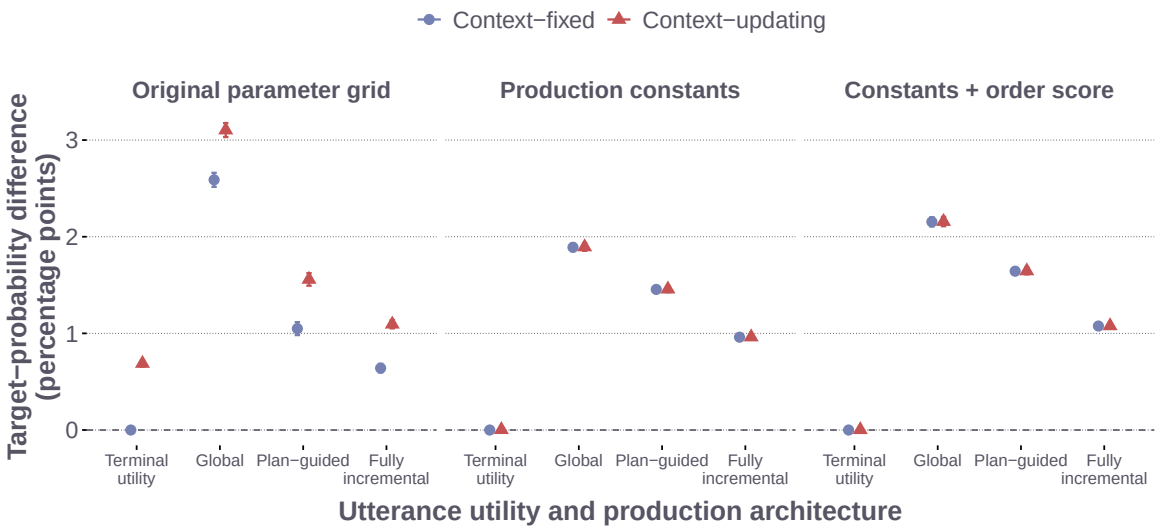


Figure 18: Controlled fifteen-response comparison across utility rules, semantic parameters, and a shared stable-order score. The terminal utility uses only the final literal target probability. The production models accumulate prefix scores: $\kappa = 0$ gives global production, $\kappa = .5$ plan-guided production, and $\kappa = 1$ fully incremental production. Panels use the original semantic grid, the production semantic constants, and those constants with the stable-order score added. Points and intervals show mean target-probability differences and ± 1.96 scene-level Monte Carlo standard errors. All comparisons use the same random scenes.

G.2 Postnominal Variant of the Controlled Simulation

The postnominal variant reuses the scenes, semantic settings, and utility rules of Table 10. Only the direction of composition changes: the noun-adjacent adjective is the first adjective of the utterance, so sequential context updating proceeds in surface order. The prefix utility follows the surface order in both versions. Context-fixed meanings do not depend on the direction of composition, so the context-fixed column applies to both versions.

Table 11: Mean size-first preference in pragmatic target recovery in percentage points for prenominal and postnominal composition on the same 24,000 random scenes.

Utility	Responses	Semantic settings	Context-fixed	Updating, prenominal	Updating, postnominal
Terminal	2	Parameter grid	0.000	1.141	−1.141
Prefix ($\kappa = 0$)	2	Parameter grid	−2.718	−1.838	−3.603
Prefix ($\kappa = 0$)	4	Parameter grid	2.257	2.772	1.830
Prefix ($\kappa = 0$)	15	Parameter grid	2.589	3.104	2.156
Prefix ($\kappa = 0$)	15	Production constants	1.891	1.894	1.887
Prefix ($\kappa = .5$)	15	Production constants	1.455	1.457	1.452
Prefix ($\kappa = 1$)	15	Production constants	0.960	0.962	0.959

Under the terminal utility, the postnominal preference is the mirror image of the prenominal one. In the fifteen-response space, updating lowers the size-first preference in the postnominal version and raises it in the prenominal version by similar amounts. At the production semantic constants, updating changes the preference by less than 0.01 percentage points in either version. Monte Carlo 95% intervals are at most ± 0.17 percentage points. The scenes are regenerated from the seed of the original sweep by `simulation/run_postnominal_variant.py`.

G.3 Supplementary Predictions on Experimental Displays

The supplementary simulation uses the size–colour trials and all eight fitted production models. For each model, 50 evenly spaced retained draws from each of its four chains give 200 posterior parameter draws. The simulation preserves the fifteen named responses, fitted semantic constants, participant parameters, stable-order score, salience calculation, and length and selection terms. For each trial and posterior draw, each of the six objects is evaluated as the intended referent while the adjective extensions and observed task covariates remain fixed. Thus changing the candidate referent does not change which colour or form the uttered words denote.

Let $S_{it}^{(d)}(u \mid o, X)$ be the resulting complete-response probability at posterior draw d. We evaluate the speaker's size-first probability conditional on the size–colour adjective set as

$$\frac{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X)}{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X) + S_{it}^{(d)}(\mathrm{CS} \mid o^*, X)}. \tag{25}$$

Separately, the pragmatic listener normalises the full-response speaker probabilities over the six referents using a uniform prior. The listener therefore retains the fitted fifteen-response normalisation when evaluating each order. We average each quantity over the original size–colour trials within posterior draw, then report the posterior mean and central 95% interval.

Figure 19 reports the resulting speaker and listener predictions. In the plan-guided models with participant-specific κ_i, the mean size-first preference in target probability was $.0062$ under fixed semantics, 95% posterior interval $[.0047, .0077]$, and $.0058$ under updating, $[.0045, .0072]$. The corresponding mean probabilities of size-first conditional on the size–colour adjective set were $.967$ and $.966$. Both fitted regimes therefore generate a size-first preference under this counterfactual listener calculation. In the response-score decomposition, the predictive difference between these models was concentrated in adjective content and length. The two calculations describe how the models differ in predicting which utterance is produced while both assigning a listener preference for size-first order.

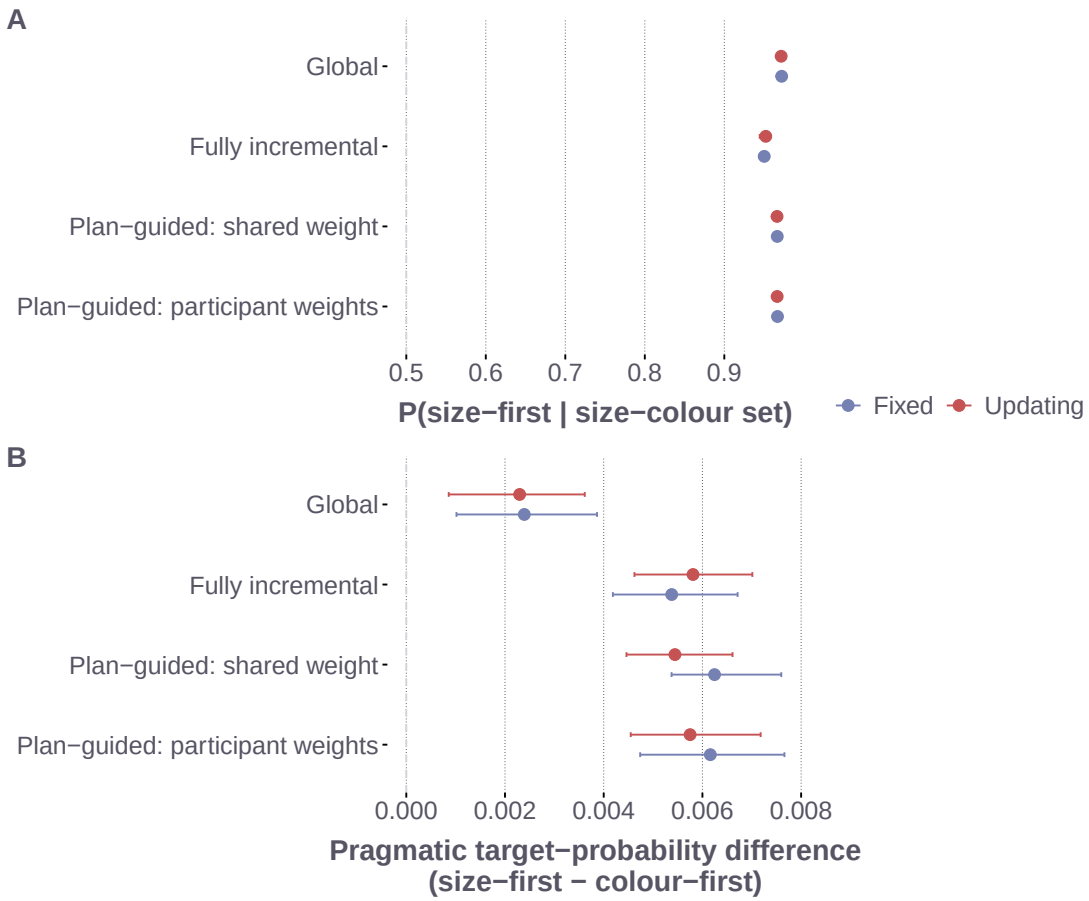


Figure 19: Supplementary predictions from the fitted production models on the size–colour displays. Panel A shows mean speaker probability of size-first conditional on selecting the size–colour adjective set. Panel B shows the difference in pragmatic target probability between size-first and colour-first. Both panels preserve the fitted fifteen-response distribution; the listener in Panel B uses a uniform prior over six referents. Intervals are central 95% posterior intervals over trial-averaged quantities using 200 posterior parameter draws per model.